eTransportation 27 (2026) 100529

Contents lists available at ScienceDirect

eTransportation

journal homepage: www.journals.elsevier.com/etransportation

Assessing decarbonization benefits of transport electrification: A provincial perspective in China

Xiwang Xiang [a,b], Tianduo Peng [a,*], Ershun Du [a,**], Chenjun Hou [a], Qin Wang [b]

[a] Laboratory of Low Carbon Energy, Tsinghua University, Beijing, 100084, China
[b] Department of Electrical and Electronic Engineering, The Hong Kong Polytechnic University, 999077, Hong Kong Special Administrative Region of China

ARTICLE INFO

Keywords:
End-use electrification
Transportation decarbonization
Provincial carbon emissions
Structural decomposition
Zero-carbon transportation

ABSTRACT

Electrification of end-use sectors is considered critical for enabling energy transition and mitigating climate change. Despite substantial expansion of electric vehicles and charging infrastructure in China, the decarbonization benefits derived from transportation electrification remain uncertain and underexplored. Here, we developed a hybrid top-down and bottom-up analytical framework integrating decomposing structural decomposition to examine emission patterns and decarbonization benefits of transportation sector across China's 30 provinces from 2000 to 2021. Our estimates reveal that (1) Economic growth and surging transport demand drove a nearly fivefold increase in carbon emissions, from 173.4 million tonnes of carbon dioxide ($MtCO_2$) in 2000 to 854.7 $MtCO_2$ in 2021. (2) Provincial electrification rates exhibit strong positive correlations with decarbonization benefits, cumulatively reducing emissions by 210.4 $MtCO_2$ (1.7 % of sectoral emissions) but with pronounced interprovincial disparities. (3) Critically, accelerated electrification generated substantial indirect carbon emissions in energy supply sectors, with lagging grid decarbonization partially offsetting direct decarbonization benefits. Consequently, we contend that electrification cannot serve as a universal policy prescription for developing economies dominated by fossil-fuel-based energy systems. Achieving just and equitable energy transitions requires synchronized regional power grids and end-use transformations. Overall, our work provides policymakers with a diagnostic toolkit to quantify electrification's decarbonization benefits, prioritize region-specific interventions, and harmonize provincial strategies with national carbon neutrality targets, offering a replicable framework for global energy transitions.

1. Introduction

The global energy system is undergoing an unprecedented transformation to mitigate climate change, with decarbonization emerging as a critical imperative across emission-intensive sectors [1] Transportation, responsible for approximately 24 % of global energy-related carbon emissions, faces formidable challenges due to its entrenched reliance on fossil fuels and fragmented infrastructure [2]. As the world's largest emitter, China's transportation sector accounted for 5.9 % of national carbon emissions in recent years, totaling 917 megatons of carbon dioxide ($MtCO_2$), and has become one of the fastest-growing contributors to the country's carbon footprint [3]. Compounding this urgency, robust evidence indicates that energy demand and emissions from China's transportation sector are projected to rise persistently, driven by sustained economic growth and rapid urbanization [4]. Consequently, decarbonizing transportation systems has become a cornerstone of China's national strategy to achieve its "dual carbon" targets (peak emissions by 2030 and carbon neutrality by 2060).

Electrification—defined as the shift from fossil fuels to electricity as the primary energy carrier—has been widely recognized as a pivotal lever for emission reduction, with numerous studies [5] affirming its critical role in sectoral decarbonization [6,7]. However, to our knowledge, few studies have quantitatively assessed the historical mitigation benefits or future decarbonization potential of electrification in key emission sectors such as transportation [8]. While recent literature [9] emphasizes the relative importance of demand-side mitigation strategies, these efforts predominantly rely on scenario modeling [10] or lifecycle assessments [11], which risk biased estimates due to overlapping effects from concurrent drivers (e.g., industrial restructuring, energy intensity improvements, and behavioral shifts) [12]. Furthermore, existing research largely focuses on long-term global pathways

* Corresponding author.
** Corresponding author.
E-mail addresses: ptd@mail.tsinghua.edu.cn (T. Peng), duershun@tsinghua.edu.cn (E. Du).

https://doi.org/10.1016/j.etran.2025.100529
Received 22 July 2025; Received in revised form 27 October 2025; Accepted 16 December 2025
Available online 19 December 2025
2590-1168/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Abbreviations

DSD – Decomposing structural decomposition
GDP – Gross domestic product
$GtCO_2$ – Gigatons of CO_2
$kgCO_2$ – Kilograms of CO_2
$MtCO_2$ – Megatons of CO_2

Symbols

C – Total carbon emissions from the transportation sector
C_j – Transportation carbon emissions across final energy types
CE_j – Transportation emission factors across final energy types
CTE – Electrification-driven decarbonization
E – Total energy consumption in the transportation sector
E_j – Transportation energy consumption across final energy types
$E\ T$ – Energy intensity in the transportation sector
EE_j – Transportation energy use share across final energy types
GP – Gross domestic product per capita
POP – Population
TAC – Value added of transportation industry
TG – Transportation demand intensity

[13] or national trajectories [14], neglecting subnational heterogeneity. In reality, the decarbonization efficacy of electrification varies markedly across China's provinces, influenced by uneven grid decarbonization, charging infrastructure gaps, and divergent consumer adoption behaviors [15]. Addressing these disparities is essential to aligning provincial actions with national targets while ensuring equity and feasibility in the energy transition. Our study seeks to address three pivotal questions:

- What are the carbon emission patterns and drivers of China's transportation sector?
- How do electrification and grid decarbonization influence transportation carbon emissions?
- How has electrification-driven historical decarbonization performed, and how to accelerate the transportation carbon transition?

To bridge these gaps, our work integrates an extended Decomposing Structural Decomposition (DSD) approach to develop a hybrid top-down and bottom-up framework for evaluating electrification-driven decarbonization effectiveness. Specifically, we first compile a provincial-level energy consumption and emission inventory (2000–2021) for China's transportation sector, disaggregated by fuel type (e.g., oil, natural gas, electricity). The analysis then quantifies the impacts of electrification and associated grid decarbonization on transportation emissions by examining socioeconomic dynamics, transport demand fluctuations, shifts in energy-specific shares, and technological advancements. Building on multi-layered decomposition results, the framework constructs an electrification-driven decarbonization assessment model to evaluate historical performance and untapped potential across three emission metrics: total decarbonization, decarbonization per capita, and decarbonization intensity. Numerical experiments are conducted to test model uncertainties, while a systematic review of existing transportation decarbonization policies and grid decarbonization initiatives informs forward-looking pathways for practical implementation.

Our study's primary contribution lies in providing a diagnostic toolkit for major emitting economies and end-use sectors to identify the efficacy of electrification-driven decarbonization. The toolkit leverages the multi-layered structural effect decomposition inherent to the DSD method, quantifying decarbonization performance driven by transportation electrification through three interconnected mechanisms: electricity substitution effect, electricity demand growth effect, and grid decarbonization effect. It marks the first attempt to evaluate and compare provincial-level mitigation benefits of electrification in China's transportation sector, offering a historical benchmark and technical roadmap to prioritize investments, tailor incentives, and harmonize interprovincial carbon quota allocation. By elucidating the synergies and bottlenecks of electrification, this work advances actionable pathways for achieving early peaking and carbon neutrality in China's transportation systems.

The subsequent sections are organized as follows: Section 2 reviews and synthesizes existing literature. Section 3 introduces the research materials and methodological framework. Section 4 demonstrates the carbon emission accounting results for China's transportation sector and highlights the decarbonization effects attributed to electrification. Section 5 comprises three subsections: Section 5.1 assesses the historical performance and future potential of decarbonization driven by transport electrification at the provincial level; Section 5.2 discusses uncertainties inherent to the model framework; Section 5.3 formulates sector-specific deep decarbonization pathways aligned with carbon neutrality objectives. At last, Section 6 synthesizes pivotal conclusions and identifies emergent research priorities.

2. Literature review

2.1. Overview of electrification transition in China's transport sector

The widespread adoption of electrification across buildings, industry, and transportation is a crucial component of global decarbonization efforts [16,17]. In the transportation sector, even if the average efficiency of internal combustion engines were to double, and the oil industry's most optimistic projections for oil supply in 2035 were realized, current technologies would still fall short of meeting the International Energy Agency's decarbonization targets [18]. The potential for decarbonization through electrification in transportation is substantial, and extensive literature now explores its effects at both national and sectoral levels, as well as its cost-effectiveness [19,20].

However, national-level analyses often obscure critical regional dynamics. While subnational assessments of transport emissions have been conducted for specific regions, such as individual states in the U.S. or major European cities, a systematic, long-term comparative analysis across the diverse provinces of a large developing country like China remains a significant research gap. This gap is particularly problematic because the decarbonization efficacy of transport electrification is not absolute but is fundamentally contingent on the carbon intensity of the regional power grid. Previous research indicates that the climate benefits of electrification are not guaranteed and are highly dependent on the cleanliness of the electricity grid, a concept often termed the transport-power system coupling effect [21,22]. This interdependence leads to profound regional heterogeneity, particularly in a country like China, where the emission reduction from an electric vehicle in a region with a hydropower-dominated grid differs vastly from one in a region reliant on coal-fired power. This provincial-level heterogeneity, driven by disparities in energy mix, economic structure, and policy environments [23], is the central analytical challenge. Yet, the historical absence of consistent provincial-level energy and emission data has confined most investigations to national or aggregated regional scales, masking these critical subnational dynamics [24,25].

2.2. Modeling strategies for transport decarbonization assessment

Carbon emission accounting models for the transportation sector can be broadly classified into two categories: top-down and bottom-up approaches. Top-down models calculate emissions based on aggregated, macro-level data. The most prominent example is the sectoral approach

recommended by the Intergovernmental Panel on Climate Change (IPCC), which calculates emissions by multiplying sectoral final energy consumption data from official statistics with corresponding emission factors [26,27]. This method is the standard for compiling national and provincial greenhouse gas inventories due to its authority and consistency. Another top-down method is the Input-Output (I-O) analysis, which can trace emissions embodied in inter-sectoral supply chains but is less suited for analyzing the direct impact of specific technological shifts within a single sector [28,29]. In contrast, bottom-up models build an emissions inventory from disaggregated, micro-level activity data. This category is diverse and includes Life Cycle Assessment (LCA) as a key framework [30]. Process-based LCA, for example, meticulously tracks emissions from every individual process in a product's or service's life cycle. While granular, it is highly data-intensive and often suffers from truncation errors by arbitrarily cutting off the system boundary. Beyond LCA, other bottom-up models focus specifically on vehicle operation, such as Europe's COPERT model (based on statistics like vehicle stock and mileage) [31] and the US EPA's MOVES model (based on real-world operational data from sources like Portable Emissions Measurement Systems) [32].

While these detailed bottom-up and LCA-based models provide high resolution and are invaluable for specific policy or product assessments, their stringent and often inconsistent data requirements make them unsuitable for a long-term, multi-province historical analysis in China, where such comprehensive and harmonized subnational data are unavailable. Therefore, this study adopts the IPCC-recommended top-down accounting method. This choice is predicated on its direct alignment with China's official provincial energy statistical system, ensuring the authority, consistency, and traceability of our foundational emission inventory. Once the inventory is established, decomposition analysis serves as a powerful tool to attribute emission changes to their underlying drivers. While methods like the Logarithmic Mean Divisia Index (LMDI) are common [33], they struggle with certain structural interdependencies. To overcome these limitations, recent advancements have led to more robust algorithms like the Generalized Divisia Index Method (GDIM) [34] and, most recently, the DSD method [35]. A key distinction between the DSD method and traditional approaches like LMDI lies in their treatment of interdependent variables, such as structural shares. Standard LMDI assumes all factors are mathematically independent, which can lead to a "structural change paradox" when applied to energy mix shares that must sum to unity, potentially yielding counterintuitive results. The DSD method is specifically designed to overcome this issue by evaluating the change in one share relative to all others, ensuring economically intuitive results [36,37]. Crucially, its capacity for multi-layered decomposition allows this study to disaggregate the effects of electrification into three distinct mechanisms: electricity substitution, electricity demand growth, and grid decarbonization—granular insights that are typically obscured in conventional models [38].

2.3. Literature gaps

Based on the foregoing review, two areas in the literature warrant further investigation. First, while the importance of the transport-power system coupling is widely acknowledged, few studies have quantitatively assessed the historical decarbonization benefits of electrification across China's diverse provinces over a long-term horizon. Such national-level analyses may obscure the subnational variations essential for effective policymaking. Second, from a methodological standpoint, while various decomposition methods are available, applying the advanced DSD method to a provincially disaggregated, top-down emission inventory for China's transport sector offers an opportunity to gain more robust and granular insights into the mechanisms of electrification.

To address these points, this study makes several contributions. Firstly, this study proposes an analytical framework that integrates a top-down emission inventory with the bottom-up decomposition technique. A key feature of this framework is its ability to leverage the multi-layered decomposition inherent to the DSD method. This allows for the quantification of decarbonization benefits from transportation electrification through three distinct, interconnected mechanisms: the electricity substitution effect, the electricity demand growth effect, and the grid decarbonization effect. This approach provides a level of detail into the drivers of electrification that is not readily available from traditional decomposition models.

Secondly, this work evaluates and compares the historical performance and future potential of transport electrification across China's 30 provinces over two decades. This analysis establishes a historical benchmark and provides a quantitative basis for policymakers. The findings can inform the tailoring of region-specific incentives, the prioritization of infrastructure investments, and the harmonization of inter-provincial carbon policies. In doing so, this study helps to connect national climate objectives with provincial implementation, offering a replicable framework for assessing end-use electrification in other major developing economies.

3. Methodologies and materials

Our work proposed an integrated assessment framework combining bottom-up and top-down approaches to analyze and compare carbon emission patterns in China's provincial transportation sectors and the decarbonization benefits brought by electrification. The framework comprises the following components: First, Section 3.1 introduces a top-down carbon accounting methodology to establish energy consumption and carbon emission inventories for provincial transportation sectors. Subsequently, Section 3.2 proposes a bottom-up carbon emission characterization model, integrating the DSD algorithm to quantify the contributions of underlying drivers and identify the roles of various energy sources in transitioning toward low-carbon transportation. Section 3.3 then develops an electrification-driven decarbonization assessment model based on decomposition results. Lastly, Section 3.4 delineates the raw data and sources used in the model framework. Fig. 1 provides a conceptual overview of our research design, illustrating the core research questions, the integrated methodological framework, and the expected outcomes, which will be elaborated in the subsequent sections.

3.1. Carbon emissions accounting model

Our study conducts top-down carbon emission accounting following the methodology provided by the Intergovernmental Panel on Climate Change. First, energy consumption within the transportation sector is uniformly converted into standard coal equivalent across energy types, then aggregated within an energy balance framework. The conversion formula is expressed as:

$$E = \sum_{i=1}^{n} E_i \times \delta_i \tag{1}$$

where E denotes the total energy consumption in standard coal equivalent, E_i is the consumption of the ith energy source, δ_i is the converted standard coal coefficient of the i-th energy source. Our study primarily focuses on five categories of final energy consumption: coal, oil, natural

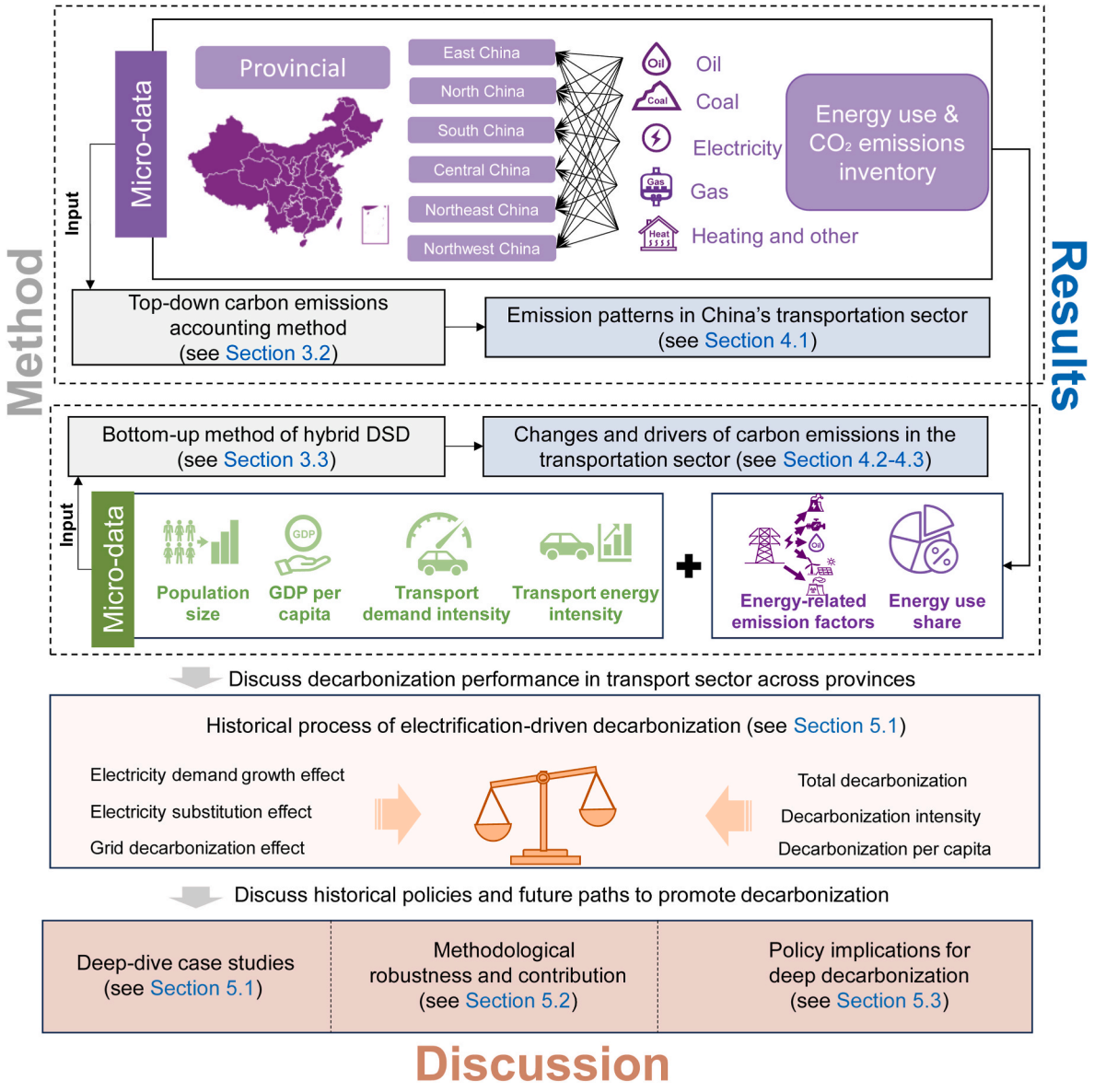


Fig. 1. Conceptual overview of the research design and workflow.

gas, heat, and electricity. Among them, coal products include raw coal, washed coal, other washed coal, type coal, coke, coke oven gas, converter gas, other gas, and other coking products. Oil products include crude oil, gasoline, kerosene, diesel, liquefied petroleum gas and fuel oil. The carbon emission accounting formula is thus defined as:

$$C = \sum_{i=1}^{n} C_i = \sum_{i=1}^{n} E_i \times f_i \times (1 - CS_i) \quad (2)$$

where C represents the transportation sector carbon emissions; C_i represents the carbon emission factor for energy source i in the transportation sector; E_i represents the consumption of the i-th energy source in the transport sector; and CS_i represents the unoxidized portion of the i-th energy source as a raw material. Since fuels used as raw materials are not directly consumed and are excluded from the total energy consumption in this work, the CS_i is considered zero when processing the data. f_i represents the carbon emission factor of for i energy source in the transportation sector, which is calculated as shown below:

$$f_i = NCV_i \times M_i \times O_i \times (44 / 12) \quad (3)$$

where the symbol NCV_i represents the average lower calorific value of energy source i; O_i denotes the oxidation rate during combustion of energy source i; M_i denotes the carbon content per unit of calorific value, the mass of carbon per unit of calorific value of the fossil energy source; and O_i and M_i denote the molecular weight of carbon dioxide and elemental carbon, respectively. Additionally, the ratio 44/12 corresponds to the molecular weight ratio of carbon dioxide to carbon.

Due to technological constraints, carbon emission factors for most energy types exhibit minimal short-term variations. Given the macro-level focus of our study, annual fluctuations in carbon emission factors for non-electric and non-thermal energy types (e.g., coal, petroleum) are

considered negligible. However, electricity and heat generated from these energy sources incur indirect carbon emissions, with their carbon emission factors significantly influenced by annual changes in regional heating/power generation mixes, grid structures, and production technologies. To ensure methodological rigor and data accuracy, our study dynamically calculates provincial carbon emission factors for electricity and heat using energy consumption statistics from 2000 to 2021. The methodology is formalized as:

$$R_t = \sum C_t \Big/ \sum E_t \tag{4}$$

Where R_t represents the carbon emission factor of electricity or heat in year t; C_t represents the total carbon emissions from the production of electricity or heat in year t; and E_t represents the total energy invested in the production of electricity/heat in year t.

3.2. Carbon emission model for transportation sector

According to the carbon accounting results in the previous section, the carbon emissions from our transportation sector can be described as:

$$C = \sum_{j=1}^{5} C_j = C_{Coal} + C_{Oil} + C_{Naturalgas} + C_{Electricity} + C_{Heat} \tag{5}$$

The symbol C represents the total carbon emissions from the transportation sector. To thoroughly examine the characteristics of carbon emissions generated by different energy sources and to explore the interrelationships among various influencing factors, this study develops a relationship model with the following characteristics:

$$C_j = POP \cdot \frac{GDP}{POP} \cdot \frac{TAC}{GDP} \cdot \frac{E}{TAC} \cdot \frac{E_j}{E} \cdot \frac{C_j}{E_j} \tag{6}$$

Eq. (6) can be simplified for visualization purposes as follows:

$$C_j = POP \cdot GP \cdot TG \cdot ET \cdot EE_j \cdot CE_j \tag{7}$$

where POP represents the population size; $GP = \frac{GDP}{POP}$ represents GDP per capita; $TG = \frac{TAC}{GDP}$ means the transportation demand intensity; $ET = \frac{E}{TAC}$ represents the transport energy intensity; $EE_j = \frac{E_j}{E}$ describes the share of the energy type i in the total energy consumption, and $CE_j = \frac{C_j}{E_j}$ represents the emission factor associated with the energy type i. Substituting Eq. (7) into Eq. (5), one obtains:

$$C = \sum_{j=1}^{5} POP \cdot GP \cdot TG \cdot ET \cdot EE_j \cdot CE_j \tag{8}$$

Leveraging the preceding carbon accounting outcomes, the DSD method is subsequently implemented for emission model decomposition. Per the DSD computational architecture [35]), Eq. (8) is formulated as a complete differential equation:

$$\begin{aligned} \mathrm{d}C = {} & \frac{\partial C}{\partial POP}\mathrm{d}POP + \frac{\partial C}{\partial GP}\mathrm{d}GP + \frac{\partial C}{\partial TG}\mathrm{d}TG + \frac{\partial C}{\partial ET}\mathrm{d}ET \\ & + \sum_{j=1}^{5}\left(\frac{\partial C_j}{\partial EE_j}\mathrm{d}EE_j + \frac{\partial C_j}{\partial CE_j}\mathrm{d}CE_j\right) \end{aligned} \tag{9}$$

A key challenge in decomposing structural effects, such as changes in energy shares (EE_j), is that these shares are interdependent and must always sum to one ($\sum_{j=1}^{5}\mathrm{d}EE_j = 0$). Traditional decomposition methods like LMDI often struggle to handle this unit-sum constraint without introducing aggregation errors or obscuring the effects of individual components. To overcome this, the DSD method introduces relaxation (F) and displacement (F_i) variables to explicitly model these interdependencies. This allows for a precise and complete decomposition without a residual term. The relationship is defined as:

$$\mathrm{d}EE_j = \mathrm{d}F_i + \mathrm{d}F \tag{10}$$

The model, as depicted by Eq. (10) can be reformulated in matrix notation, and is detailed as follows:

$$\begin{bmatrix} 1 & -\frac{\partial C_1}{\partial EE_1} & -\frac{\partial C_2}{\partial EE_2} & -\frac{\partial C_3}{\partial EE_3} & -\frac{\partial C_4}{\partial EE_4} & -\frac{\partial C_5}{\partial EE_5} & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathrm{d}C \\ \mathrm{d}EE_1 \\ \mathrm{d}EE_2 \\ \mathrm{d}EE_3 \\ \mathrm{d}EE_4 \\ \mathrm{d}EE_5 \\ \mathrm{d}F \end{bmatrix} =$$

$$\begin{bmatrix} \frac{\partial C}{\partial POP} & \frac{\partial C}{\partial GP} & \frac{\partial C}{\partial TP} & \frac{\partial C}{\partial ET} & \frac{\partial C_1}{\partial EE_1} & \cdots & \frac{\partial C_5}{\partial EE_5} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathrm{d}POP \\ \mathrm{d}GP \\ \mathrm{d}TP \\ \mathrm{d}ET \\ \mathrm{d}EE_1 \\ \vdots \\ \mathrm{d}EE_5 \\ \mathrm{d}CE_1 \\ \vdots \\ \mathrm{d}CE_5 \end{bmatrix} \tag{11}$$

Thereafter, we denote endogenous variables by $\mathrm{d}\alpha = [\mathrm{d}C, \mathrm{d}EE_1, \cdots \mathrm{d}EE_5, \mathrm{d}F]^{\mathrm{T}}$ and exogenous variables by $\mathrm{d}\beta = [\mathrm{d}POP, \mathrm{d}GP, \mathrm{d}TP, \mathrm{d}ET, \mathrm{d}EE_1, \cdots \mathrm{d}EE_5, \mathrm{d}CE_1, \cdots \mathrm{d}CE_5]^{\mathrm{T}}$, and set $\Gamma = \gamma(\alpha, \beta)$, $\mathrm{I} = \delta(\alpha, \beta)$. Eq. (11) can be rewritten by:

$$\Gamma \cdot \mathrm{d}\alpha = \mathrm{I} \cdot \mathrm{d}\beta \tag{12}$$

Through mathematical simplification yields:

$$\mathrm{d}\alpha = \Gamma^{-1} \cdot \mathrm{I} \cdot \mathrm{diag}(\mathrm{d}\beta) \cdot \varepsilon \tag{13}$$

The term $\mathrm{diag}(\mathrm{d}\beta)$ signifies a diagonal matrix, while ε represents a vector of ones. Subsequently, numerical integration methods are employed to solve Eq. (13) yielding the results.

$$\begin{cases} \prod^{(n)} = \left(\Gamma^{(n-1)}\right)^{-1} \cdot I^{(n-1)} \cdot \mathrm{diag}(\mathrm{d}\beta) \\ \mathrm{d}\alpha^{(n-1)} = \prod^{(n)} \cdot \varepsilon \\ \beta^{(n)} = \beta^{(n-1)} + \mathrm{d}\beta \\ \alpha^{(n)} = \alpha^{(n-1)} + \mathrm{d}\alpha^{(n-1)} \\ \Gamma^{(n)} = \gamma(\alpha^{(n)}, \beta^{(n)}) \\ \mathrm{I}^{(n)} = \delta(\alpha^{(n)}, \beta^{(n)}) \end{cases} \tag{14}$$

The final decomposition result is a sum accumulated through an iterative process by successively adding intervals one by one.

$$\prod = \sum_{n=1}^{N} \prod^{(n)} \tag{15}$$

Thus, the role of potential drivers of the endogenous vector of carbon emissions change (ΔC) can be expressed as follows:

$$\Delta C = \Delta POP + \Delta GP + \Delta TG + \Delta ET + \Delta EE + \Delta CE \tag{16}$$

Changes in population size, economic activity, transportation activity, and energy intensity are represented by the variables ΔPOP, ΔGP, ΔTG, and ΔET, respectively. Additionally, the combined effects of energy emission factors and energy structure are represented by ΔCE and ΔEE, which can be further disaggregated by energy type.

$$\begin{cases} \Delta EE = \Delta EE_{Coal} + \Delta EE_{Oil} + \Delta EE_{Natura\ lg\ as} + \Delta EE_{Electricity} + \Delta EE_{Heat} \\ \Delta CE = \Delta CE_{Coal} + \Delta CE_{Oil} + \Delta CE_{Natura\ lg\ as} + \Delta CE_{Electricity} + \Delta CE_{Heat} \end{cases} \quad (17)$$

In Eq. (17), the expressions on the right-hand side reveal the direct impact of energy use proportions and source-specific emission factors on transportation carbon emissions.

3.3. Electrification-driven decarbonization assessment

To evaluate the decarbonization benefits of provincial transportation electrification, this section constructs an electrification-driven decarbonization assessment model based on the aforementioned multi-layer decomposition results. The unique diagnostic power of our framework lies in its ability to precisely isolate the distinct mechanisms through which electrification impacts emissions. The energy types in the transportation sector are categorized into the electricity part and the non-electricity part (including coal, oil, natural gas and heat). The proposed model can be reformulated as follows:

$$C = POP \cdot \frac{GDP}{POP} \cdot \frac{TAC}{GDP} \cdot \frac{E}{TAC} \cdot \left(\frac{E_{Electricity}}{E} \cdot \frac{C_{Electricity}}{E_{Electricity}} + \frac{E_{Non-Electricity}}{E} \cdot \frac{C_{Non-Electricity}}{E_{Non-Electricity}} \right) \quad (18)$$

Eq. (18) can be simplified as:

$$C = POP \cdot GP \cdot TG \cdot ET \cdot \left(EE_{Electricity} \cdot CE_{Electricity} + EE_{Non-electricity} \cdot CE_{Non-electricity} \right) \quad (19)$$

Based on the decomposition results from Eq. (17), we can isolate the terms related to the structural shift towards electricity and the changing carbon intensity of the grid. The total decarbonization benefit of transportation electrification (*CTE*) is therefore defined as the sum of three distinct effects derived directly from our DSD analysis:

$$CTE = \Delta EE_{Non-electricity} + \Delta CE_{Electricity} + \Delta EE_{Electricity} \quad (20)$$

Where each component represents a specific decarbonization mechanism: The first component, the electricity substitution effect ($\Delta EE_{Non-electricity}$), quantifies the direct emission reductions achieved by displacing carbon-intensive fossil fuels such as oil, coal, and gas with electricity, where a negative value indicates a net decrease in emissions due to the declining share of non-electric energy sources. The second, the grid decarbonization effect ($\Delta CE_{Electricity}$), captures the impact of changes in the carbon intensity of electricity generation; a negative value here reflects a cleaner power grid—for instance, through the expansion of renewable energy—which enhances the climate benefit associated with each unit of electricity consumed by the transport sector. Finally, the electricity demand growth effect ($\Delta CE_{Electricity}$) measures the consequences of the expanding share of electricity in the total transport energy mix, thereby accounting for the scale of electrification itself. This structured framework enables a diagnostic analysis that moves beyond simple emissions accounting. It allows us to identify, for example, why electrification may underperform in certain provinces—such as when a strong substitution effect is offset by an insufficient grid decarbonization effect, resulting in minimal net climate benefit. Such detailed insight is essential for designing targeted and effective regional decarbonization policies.

3.4. Study area and data

The analytical unit of this study comprises 30 provincial-level administrative regions in China, excluding Tibet, Hong Kong, Macao, and Taiwan due to data availability and statistical consistency issues. A provincial-level perspective was chosen because energy and transport policies in China are largely formulated and implemented at the subnational level, and more critically, because significant heterogeneity exists among provinces in terms of economic structure, population size, energy endowment, and transport demand. A national-level macroeconomic analysis would average out these crucial differences, thereby masking the true performance of electrification-driven emission reductions across diverse regions. As summarized in Table 1, the descriptive statistics for the key variables in our study clearly reveal this substantial inter-provincial disparity. For instance, the standard deviation for annual transport carbon emissions across provinces is as high as 21.28 $MtCO_2$, and the maximum value is over 150 times greater than the minimum. This pronounced heterogeneity fundamentally warrants a subnational analytical approach, enabling the provision of a granular scientific basis for tailoring region-specific decarbonization strategies. Regarding the emission scope and system boundaries, our accounting framework adopts a tank-to-wheel scope for direct fossil fuel combustion and encompasses indirect emissions from purchased electricity. The analysis is constrained by the primary data source, which consolidates provincial energy consumption and emissions into a single "Transport, Storage, and Post" sector.

The data required for our modeling were sourced from official statistical publications. Energy consumption data for the transport sector in each province, covering five main categories (coal, oil, natural gas, electricity, and heat), were primarily extracted from the regional energy balance sheets in the *China Energy Statistical Yearbook* series (https://data.stats.gov.cn/english). For details on the regional grid divisions relevant to transport decarbonization, please refer to Supplementary Material A. The conversion coefficients for carbon emission accounting were mainly referenced from the *General Principles for Calculation of Comprehensive Energy Consumption* (https://www.chineses-tandard.net) and the *Guidelines for the Compilation of Provincial Greenhouse Gas Inventories* (see Supplementary Material B and Supplementary Material C for details). Linear interpolation was used to address missing data points. Other macroeconomic indicators, including population, Gross Domestic Product (GDP), and the value-added of the transport, storage, and postal sectors, were obtained from the official website of the *National Bureau of Statistics of China* and the *China Statistical Yearbook* series (https://www.stats.gov.cn/english/Statisticaldata/year-book/).

4. Results

This section presents the empirical findings derived from the integrated assessment framework established in Section 3. Our analysis systematically unfolds in three stages to address the core research questions. First, Section 4.1 elucidates the spatiotemporal dynamics and structural characteristics of carbon emissions within China's provincial transportation sector from 2000 to 2021, establishing a foundational emissions baseline. Subsequently, Section 4.2 employs the DSD method to dissect the contributions of key socioeconomic and technological drivers to the observed emission changes. Finally, Section 4.3 culminates in a focused analysis of the decarbonization effects attributable specifically to shifts in the energy mix and advancements in energy-related technologies, with a particular emphasis on the role of electrification.

4.1. Emission patterns in China's transportation sector

Based on the carbon accounting model, we compiled a comprehensive inventory of energy consumption and carbon emissions for China's 30 provinces from 2000 to 2021. We also compared the carbon emission accounting results with those of the China Carbon Accounting Database [39], and the comparison showed that the relative error between the two is only 5 % (see Supplementary Material C for details), and the comparison verified the reliability of the accounting results. Fig. 2 presents the total carbon emissions from transportation in China from 2000 to 2021 and their distribution among the five energy sources (coal, oil, natural gas, electricity, and heat) and the six grid regions (north China, northeast China, east China, central China, south China, and northwest China). Over this period, cumulative carbon emissions amounted to 12.8 gigatons of carbon dioxide ($GtCO_2$). Annual carbon emissions increased

Table 1
Descriptive statistics of key variables for modelling (2000–2021).

Symbol	*POP*	*GDP*	*TAC*	*E*	*C*
Variable	Population	GDP	Transportation demand	Total energy use	Total CO_2 emissions
Definition	Total population of the province	Gross domestic product	Value added of transportation industry	Total final energy consumption by transportation sector	Total CO_2 emissions from transportation sector
Unit	Million	Billion yuan	Billion yuan	Mtce	$MtCO_2$
Mean	44.55	8032.76	28.25	19.32	8.51
Std. Dev	27.31	12323.45	19.84	14.26	6.50
Max	126.84	94330.00	85.95	88.80	35.43
Min	5.17	26.37	1.75	0.60	0.23
Observations	660	660	660	660	660

Notes: The table summarizes the statistical characteristics of all annual observations for the 30 provinces over the 22-year study period. Std. Dev denotes standard deviation; Max and Min denote maximum and minimum values, respectively.

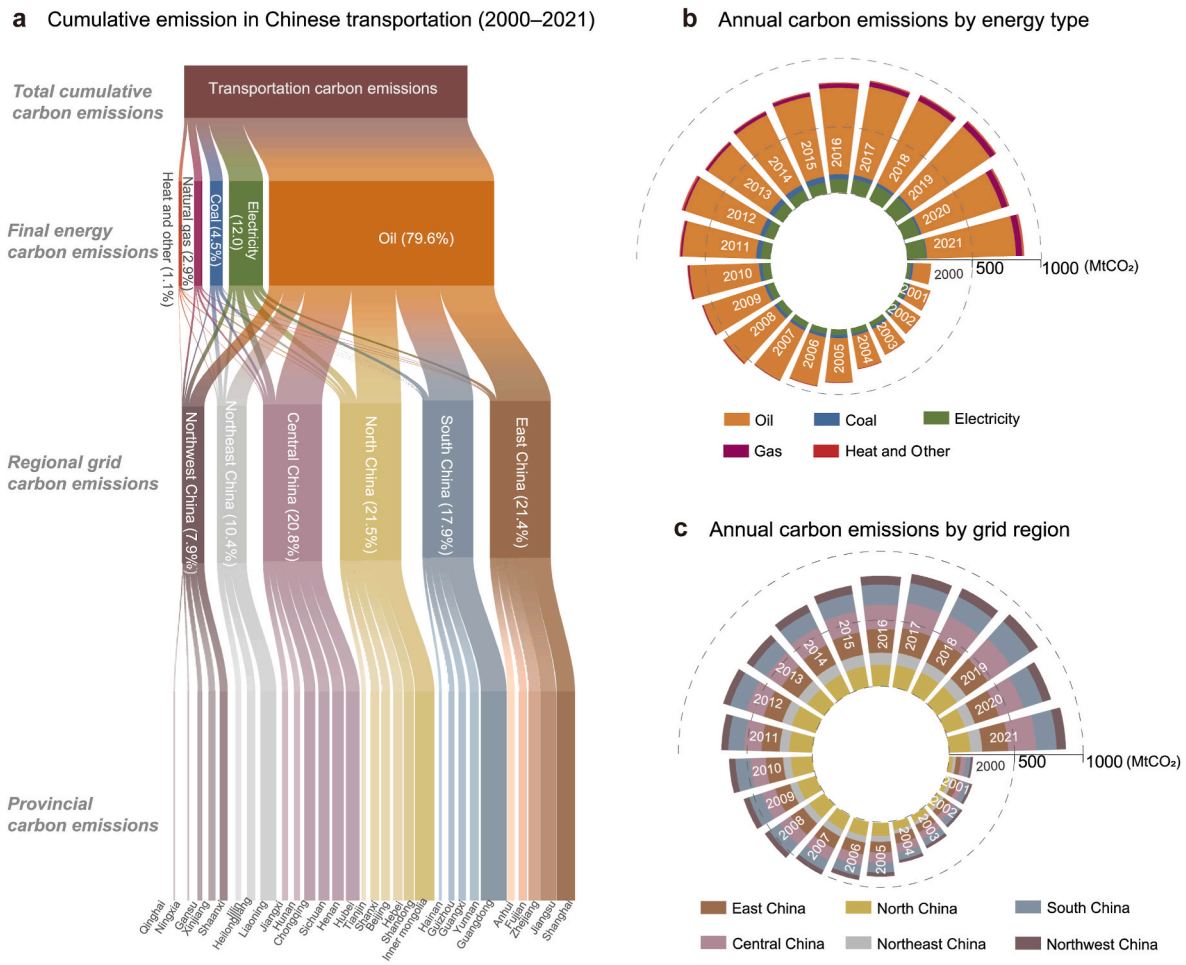


Fig. 2. National, regional, and provincial carbon emissions from China's transportation sector, 2000 to 2021: (a) Transportation emission flows from source to province in China from 2000 to 2021; (b) Transportation carbon emissions by energy type; (c) Transportation carbon emissions by grid region.

from 173.4 $MtCO_2$ in 2000 to 854.7 $MtCO_2$ in 2021, with an average annual growth rate of 8.2 %. However, a distinct inflection point is visible around 2010; the growth rate decelerated markedly from 13.5 % (2000–2010) to 3.3 % (2010–2021). This slowdown is not incidental but reflects a profound shift in China's development paradigm. It coincides with the introduction of the "New Normal" economic phase, which signaled a move away from unfettered, resource-intensive growth towards a more sustainable, high-quality development model [40]. This macroeconomic shift, coupled with increasingly stringent energy conservation and emission reduction targets for the transport sector embedded in China's 12th (2011–2015) and 13th (2016–2020) Five-Year Plans, provided the top-level policy pressure to temper emission growth.

Despite rapid electrification progress, the emission structure of China's transportation sector remained relatively stable (Fig. 2 a). Oil maintained dominance (10.1 $GtCO_2$, 79.6 %). The most notable change is the sharp decline of coal-related emissions, particularly after 2015. This is a direct consequence of national environmental campaigns, most prominently the "Air Pollution Prevention and Control Action Plan" [41], which aggressively phased out coal-fired locomotives and curbed industrial coal transport. Concurrently, emissions from electricity and natural gas consumption have steadily risen. This trend is policy-driven, resulting from national strategies to promote cleaner energy substitutes and the rapid expansion of the new energy vehicle market, which has

been heavily subsidized by the central government [42].

At the grid-region level (Fig. 2 b), the four highest-emitting regions collectively contributed 81.6 % of national totals, concentrated in China's economically dynamic eastern and southern corridors. Significant heterogeneity in emission trajectories reflects diverse developmental stages and policy priorities. For instance, the North China grid, which includes the heavily regulated Beijing-Tianjin-Hebei region, saw its emissions stabilize post-2012. This can be attributed to the intense pressure of coordinated regional air quality control measures that forced industrial restructuring and transport optimization. In contrast, emissions in the Central China grid continued to rise, peaking in 2021. This reflects the classic pattern of developing economies, where these provinces undertook large-scale industrialization and urbanization later than the coastal regions, leading to a period of "catch-up" growth in transport demand and associated emissions [43] Provincially, Guangdong recorded the highest cumulative emissions (1.16 $GtCO_2$), yet its post-2010 growth rate fell below the national average, signaling effective emission controls. Inner Mongolia emerged as the sole province achieving negative emission growth (−0.9 %) in the past decade, attributable to its accelerated adoption of clean energy in transportation.

In summary, the provincial-level energy consumption and carbon emission accounting by fuel category has established an accurate carbon emission baseline for China's transportation sector, while providing essential data support for subsequent integrated assessment modeling and results analysis, serving as a critical prerequisite for addressing the three research questions posed in Section 1.

4.2. Drivers of carbon emissions in the transportation sector

Fig. 3 analyzes the changing trends of carbon emissions in China's provincial transportation sectors over the past two decades and their underlying driving factors. Regarding factors promoting transportation carbon emissions: economic growth (measured by per capita GDP) was the primary driver of transportation carbon emission increases across provinces, contributing 2.28 $GtCO_2$, accounting for 78.2 % of emission increments. Energy intensity was the second largest contributing factor, responsible for 0.54 $GtCO_2$, representing 18.5 % of carbon emission growth. Exceptions existed, such as in Shanxi province during 2010–2021, where the energy structure pattern effect became the second largest emission driver (0.28 $GtCO_2$) after per capita GDP. Moreover, energy intensity transitioned from being an emission-promoting factor (17.3 $MtCO_2$ before 2010) to an emission-reducing factor (−0.2 $MtCO_2$ after 2010), primarily due to changes in Shanxi's transportation energy structure. Similar trends were observed in Inner Mongolia and Shandong province, which also belong to the north China regional grid.

Concerning factors promoting provincial transportation decarbonization: Transportation demand intensity was the dominant decarbonization factor, reducing emissions by 2.22 $GtCO_2$ over the past 20 years, accounting for 98.1 % of all reduction factors, indicating close correlation between transportation activities level and carbon emissions. Before 2010, energy consumption structure (−23.1 $MtCO_2$) was the second most important decarbonization factor, followed by emission factors (−10.5 $MtCO_2$). However, after 2010, transportation energy structure became an emission-increasing factor (41.3 $MtCO_2$), reflecting how rapid growth in transportation demand offset decarbonization benefits despite mitigation efforts. Notably, the impact of various driving factors on carbon emissions varied according to provincial population and economic conditions. For example: In Hainan province, energy intensity was an emission-increasing factor (4.9 $MtCO_2$) during 2000–2010 but became a decarbonization factor (−2.7 $MtCO_2$) during 2010–2021. Shanxi (from 17.3 to −0.2 $MtCO_2$), Yunnan (from 12.7 to −2.3 $MtCO_2$), and Shaanxi (from 12.5 to −2.7 $MtCO_2$) showed similar trends, mainly due to adoption of new energy technologies and improved energy efficiency, leading to reduced energy consumption per unit of transportation and lower emissions. Ningxia exhibited the opposite trend (from −0.8 to 2.8 $MtCO_2$), primarily because insufficient energy efficiency improvements led to increased energy consumption per unit of transportation with insignificant reductions in transportation energy use. Additionally, before 2010, population factors increased emissions in Heilongjiang province (0.1 $MtCO_2$), but after 2010, population factors transitioned to a decarbonization effect (−4.8 $MtCO_2$). Similar patterns were observed in Inner Mongolia and Gansu, suggesting that local transportation sector emissions no longer increased with population growth due to factors like slowing population growth, outmigration, or transportation structure optimization. It's noteworthy that due to significant differences in transportation activities and technological levels across provinces, the decarbonization effects of energy structure and energy-related emission factors are discussed in greater detail in Section 4.3.

To further elucidate the interplay between these drivers, a Pearson correlation analysis was conducted on the key parameters (see Table S4 in the Supplementary Material D). The analysis reveals a strong positive correlation between GDP per capita and total carbon emissions ($r = 0.66$, $p < 0.01$), providing statistical evidence for its role as the primary driver. Conversely, a significant negative correlation exists between GDP per capita and transport demand intensity ($r = -0.95$, $p < 0.01$), which empirically supports the observation of a decoupling trend where economic growth becomes less transport-intensive over time. This suggests that while economic prosperity drives overall demand, it also coincides with structural and efficiency improvements that temper this effect. Based on the magnitude of the cumulative contributions calculated by our DSD model over the 2000–2021 period, the relative importance of these drivers is ranked as follows: 1) Economic growth, which was the most significant factor increasing emissions; 2) Transport demand intensity, which served as the largest offsetting factor promoting decarbonization; and 3) Energy intensity, which was the second-largest contributor to emission growth. This ranking provides a clear, data-driven hierarchy of the forces shaping China's transport emissions trajectory over the last two decades.

In summary, these results reveal the changing trends of carbon emissions in China's provincial transportation sectors and the driving and restraining forces behind these changes, thereby addressing research question 1 raised in Section 1.

4.3. Decarbonization effects of transport electrification

Based on the multi-layer decomposition results of the structural decomposition algorithm, Fig. 4 provides a detailed analysis of the contributions of changes in energy mix shares (coal, electricity, oil, natural gas, and heat) and their corresponding emission factors to provincial transportation carbon emissions from 2000 to 2021. This reveals the decarbonization effects driven by shifts in provincial transportation energy structures and low-carbon technologies associated with specific energy sources.

For the decarbonization effects of technological progress (Fig. 4 a), technological innovations in oil-based systems emerged as the primary driver of emission reductions in the transport sector during the first decade of the 21st century (2000–2010), achieving a measurable decrease of 6.1 $MtCO_2$ (47.2 % of total decarbonization), followed by electricity (4.0 $MtCO_2$, 31 %) and coal (2.8 $MtCO_2$, 21.8 %). However, from 2010 to 2021, electricity (−17.2 $MtCO_2$) became the predominant decarbonization contributor, while coal and oil remained emission-increasing drivers. These shifts reflect China's efforts to restructure its energy mix and promote cleaner energy sources like electricity, underscoring the need to further reduce oil dependence and expand clean energy adoption in transportation. At the provincial level, the impacts of energy-related emission factors varied markedly. For instance, in Sichuan province, emission factors shifted from decarbonizing (−3.8 $MtCO_2$ during 2000–2010) to emission-increasing (1.7 $MtCO_2$ during 2010–2021), primarily due to a reversal in oil emission factors (−2.6 $MtCO_2$ to 2.3 $MtCO_2$). In contrast, Shandong province transitioned from emission-increasing factors (0.6 $MtCO_2$ pre-2010) to decarbonizing

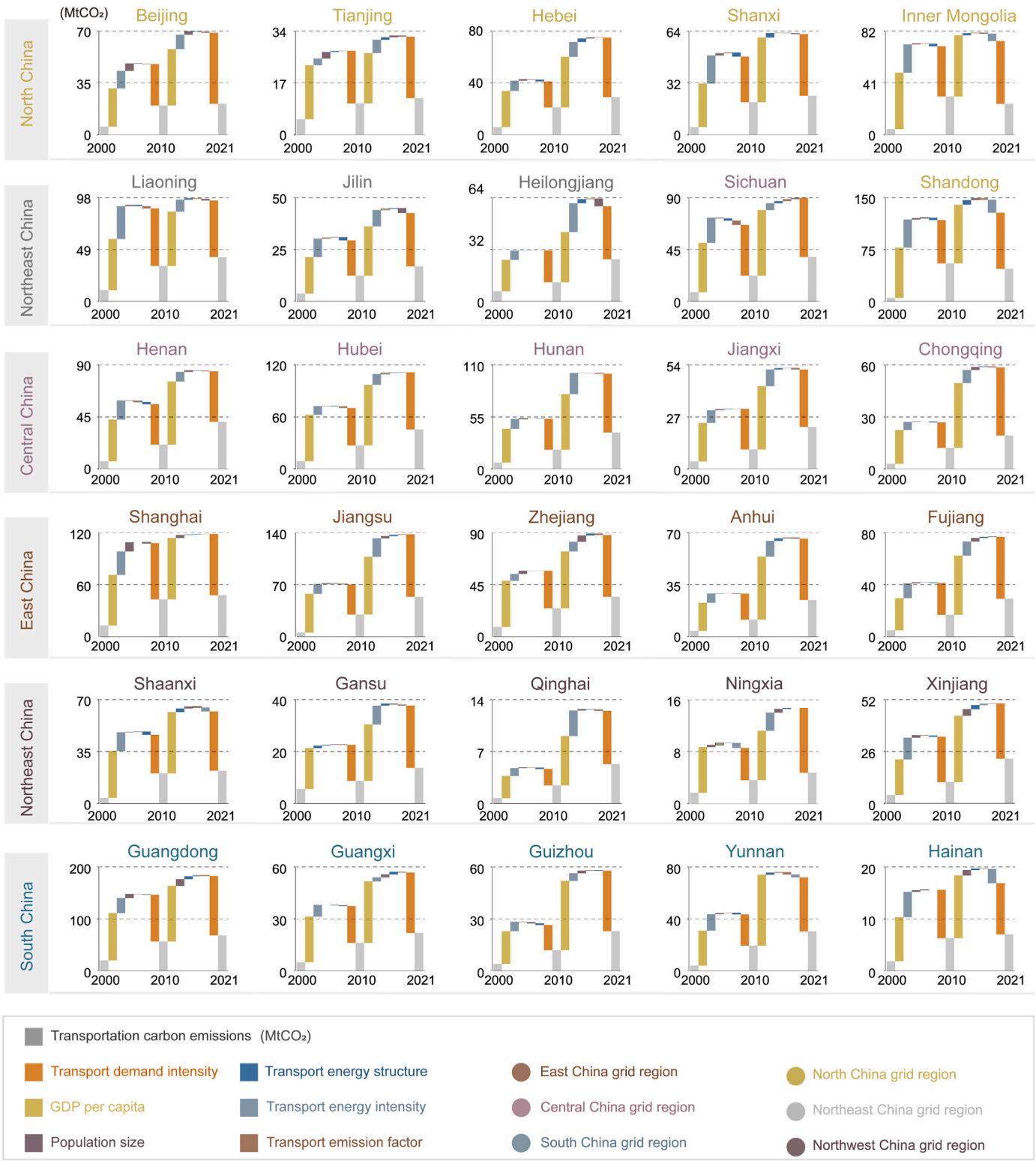


Fig. 3. Changes and drivers of carbon emissions from the transportation sector in 30 provinces of China (2010–2021). Notes: The figure illustrates the contributions of six key drivers to the change in transport carbon emissions for 30 Chinese provinces over two periods: 2000–2010 and 2010–2021. The y-axis represents total carbon emissions in $MtCO_2$. The colored segments within each bar quantify the positive (emission-increasing) or negative (emission-reducing) impact of each driver: Transport demand intensity (orange), GDP per capita (yellow), Population size (wine), Transport energy structure (dark blue), Transport energy intensity (light blue), and Transport emission factor (brown).

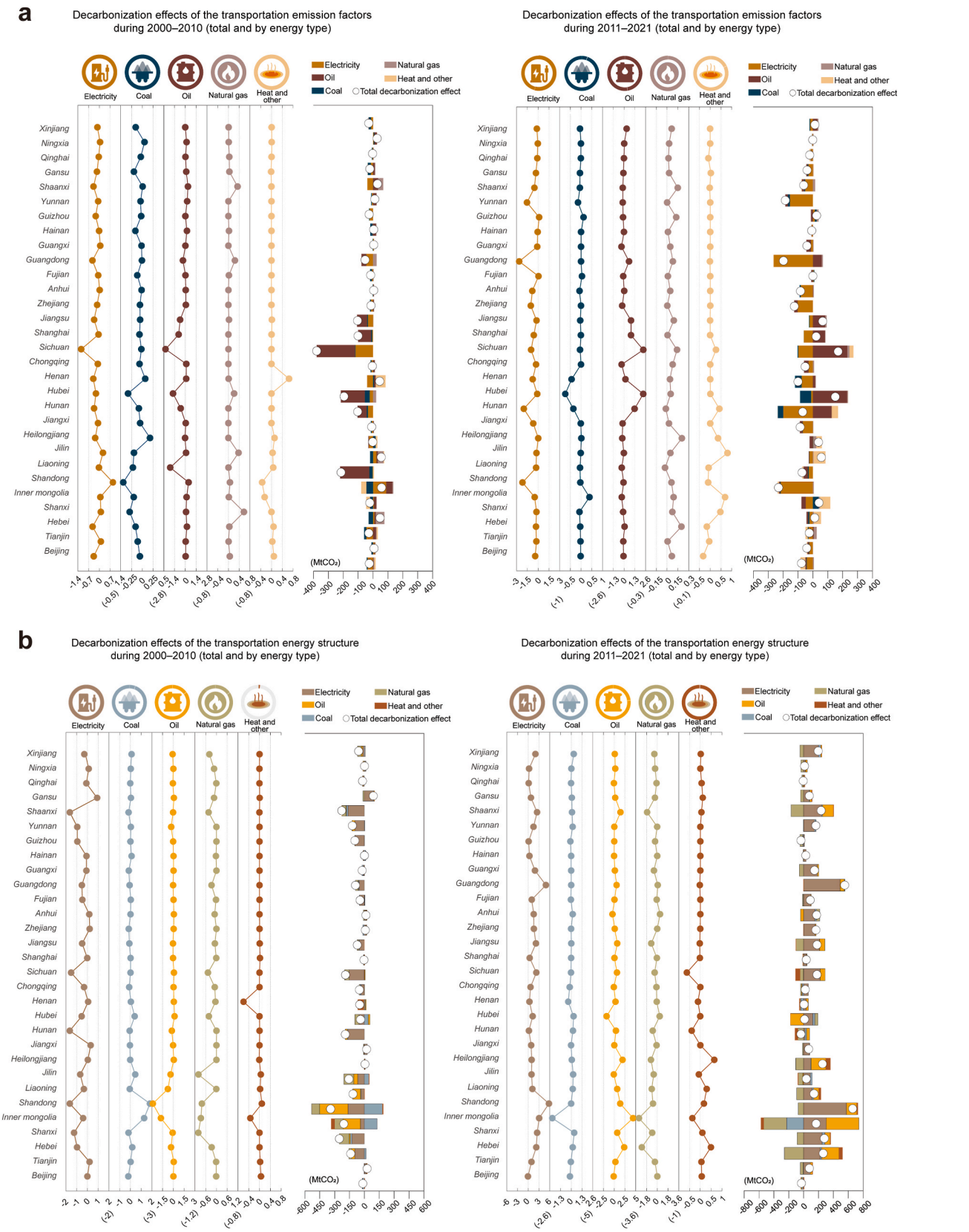


Fig. 4. Decarbonization effects of transportation energy types across Chinese provinces (total and by energy type): (a) Decarbonization effects of the transportation emission factors in each province for the periods 2000–2010 and 2010–2021; (b) Decarbonization effects of the transportation energy structure in each province for the periods 2000–2010 and 2010–2021. Notes: The x-axis measures the change in emissions in $MtCO_2$, with negative values indicating decarbonization. The y-axis lists the 30 provinces. The lines track the contribution of electricity, coal, oil, natural gas, and heat, while the bar chart on the right shows the total net effect.

effects (−2.3 $MtCO_2$ post-2010), with electricity's contribution shifting from 0.9 $MtCO_2$ to −2.2 $MtCO_2$, highlighting the growing role of electrification in sectoral decarbonization.

Regarding the decarbonization effects of energy structure transitions (Fig. 4 b), shifts in transportation energy mixes initially delivered significant decarbonization benefits across provinces, though these effects weakened post-2010 and even reversed in most regions. This reversal is attributed to changes in emission factors for electricity (from −12.7 $MtCO_2$ pre-2010 to 43.6 $MtCO_2$ post-2010) and oil (from −6.5 $MtCO_2$ to 14.2 $MtCO_2$). In contrast, natural gas exhibited stable decarbonization effects, increasing reductions from −6.7 $MtCO_2$ (2000–2010) to −16.1 $MtCO_2$ (2010–2021), highlighting its potential as a transitional energy for China's transportation decarbonization. Provincially, regions like Henan (−2.3 $MtCO_2$), Guizhou (−1.3 $MtCO_2$), and Beijing (−0.4 $MtCO_2$) maintained structural decarbonization benefits, while others that previously contributed to decarbonization shifted to emission increases post-2010. For example, Sichuan's emissions transitioned from −1.9 $MtCO_2$ to 1.8 $MtCO_2$, driven by electricity mix shifts (from −1.5 $MtCO_2$ to 2.2 $MtCO_2$), emphasizing the critical role of clean electricity. Similar trends occurred in Inner Mongolia (−2.1 $MtCO_2$ to 1.7 $MtCO_2$), Jilin (−1.6 $MtCO_2$ to 0.4 $MtCO_2$), and Chongqing (−0.5 $MtCO_2$ to 0.2 $MtCO_2$), largely due to rising transportation emissions and lagging grid decarbonization.

Collectively, these findings elucidate the decarbonization benefits derived from China's transportation energy structure transitions and energy-specific low-carbon technologies over the past two decades, thereby addressing research question 2 posed in Section 1.

5. Discussion

Building on the empirical results presented in Section 4, this section synthesizes and interprets these findings to derive deeper insights into the effectiveness and complexities of transportation electrification in China. We begin in Section 5.1 by examining the historical trajectory of electrification-driven decarbonization, critically assessing the synergies and trade-offs with the power sector's carbon emissions. To ensure the credibility of our conclusions, Section 5.2 addresses the methodological robustness of our framework through a rigorous uncertainty analysis. Finally, Section 5.3 contextualizes our findings within China's current policy landscape and culminates in the formulation of evidence-based, deep decarbonization strategies designed to accelerate the transition toward a zero-carbon transportation system.

5.1. Historical process of electrification-driven decarbonization

Fig. 5 depicts the historical trajectory of decarbonization driven by electrification in China's transportation sector and the corresponding power-related carbon emissions at the national level over the past two decades. This analysis reveals the trade-offs and synergies between transportation electrification development and additional power sector carbon emissions. Commencing in 2000, the substitution of conventional fuels with electricity in clean infrastructure, such as electric vehicles and electric buses, coupled with power system decarbonization, delivered significant direct decarbonization benefits to China's transportation sector. The total decarbonization volume exhibited an overall upward trend, with an average annual growth rate of 3.8 % and a mean annual decarbonization volume of 10.1 (±4.7) $MtCO_2$. However, it is crucial to note that periods of rapid electrification growth also induced substantial indirect carbon emissions. Power-related carbon emissions within the transportation sector surged from 26.1 $MtCO_2$ in 2001 to 136.9 $MtCO_2$ in 2021 – an increase exceeding fourfold. This phenomenon is primarily attributable to China's current coal-dominated power supply structure. In other words, when the pace of electrification accelerates significantly, the mismatched rate of power system decarbonization can partially offset or even negate the direct decarbonization benefits derived from electrification. Consequently, vigilance is warranted regarding the potential for elevated electrification levels to introduce additional emission risks. Although end-use electrification is widely regarded as a universal remedy for achieving carbon neutrality, it should not be implemented as a one-size-fits-all solution, particularly in developing nations where fossil fuels dominate the energy supply mix.

Fig. 5 illustrates the spatiotemporal evolution of electrification-driven decarbonization patterns across China's 30 provinces over the past twenty years. Cumulative decarbonization from transportation electrification reached 210.4 $MtCO_2$ during this period, equivalent to 1.7 % of the sector's concurrent carbon emissions. Aligning with national-level findings, electrification-driven decarbonization benefits generally exhibit a positive correlation with provincial electrification penetration rates. However, exceptions exist, such as in Guizhou, where the electrification rate decreased by 3.5 % between 2000 and 2010 compared to 2011 and 2021, yet the contribution to decarbonization increased by 0.3 $MtCO_2$. In Henan, despite a decrease in the electrification rate by 1.2 % over the same periods, the contribution to decarbonization increased by 7.4 $MtCO_2$. This is primarily due to improvements in the power structure, enhanced power decarbonization,

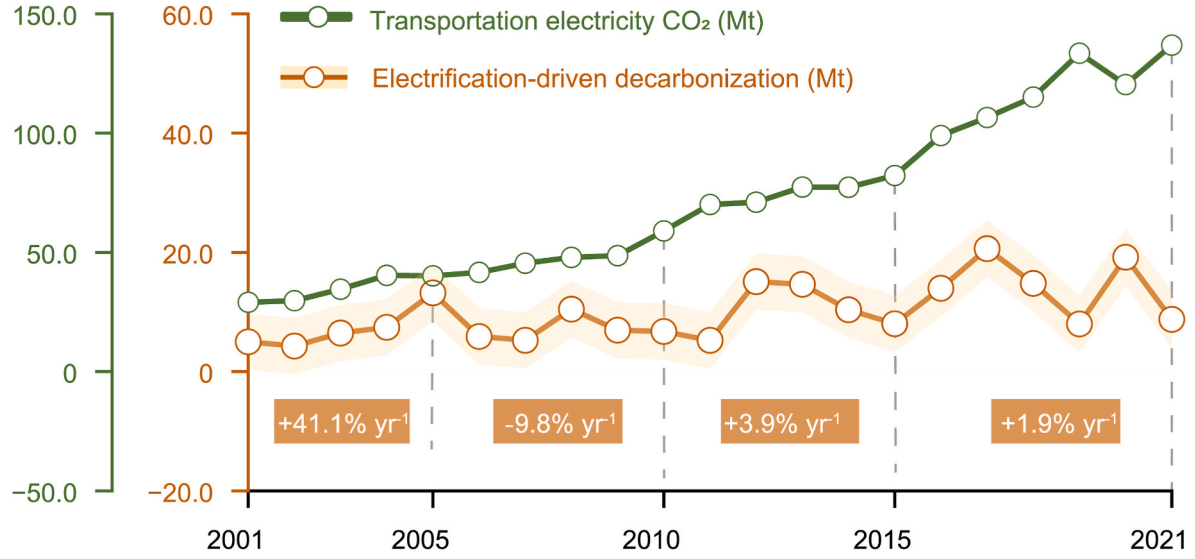


Fig. 5. Historical trajectory of electricity carbon emissions and electrification-driven decarbonization in China's transportation sector. Note: the shading in the line chart indicates one standard deviation.

and the implementation of supportive policies, highlighting the need for a comprehensive assessment of the contribution of electrification to emission reduction. At the regional level, significant carbon emission reductions through transportation electrification between 2000 and 2010 were primarily concentrated in the north China power grid (24.5 $MtCO_2$) and the central China power grid (16.1 $MtCO_2$). Before 2010, there was a trend of higher reductions in the north compared to the south. However, after 2010, decarbonization accelerated in the southern regional grid, with significant contributions to carbon emission reductions mainly from the southern regional grid (36.9 $MtCO_2$), northern regional grid (32.8 $MtCO_2$), and central China regional grid (31.9 $MtCO_2$). At the Provincial level, Guangdong province made the largest contribution to decarbonization from transport electrification, amounting to 34.8 $MtCO_2$. Despite the general increase in the contribution of electrification to decarbonization across most provinces after 2010, exceptions include Shandong province, where electrification contributed only 4 $MtCO_2$ between 2010 and 2021, and Shanxi province, where the contribution dropped from 4.2 $MtCO_2$ between 2000 and 2010 to 3 $MtCO_2$ between 2010 and 2021.

The decarbonization efficiency, as referenced from other studies, is defined in our study as the ratio of the total decarbonization to the total carbon emissions from transportation. This ratio reveals the decarbonization potential of a region's electrification development. Overall, the efficiency of the east China grid was the highest before 2010. After 2010,

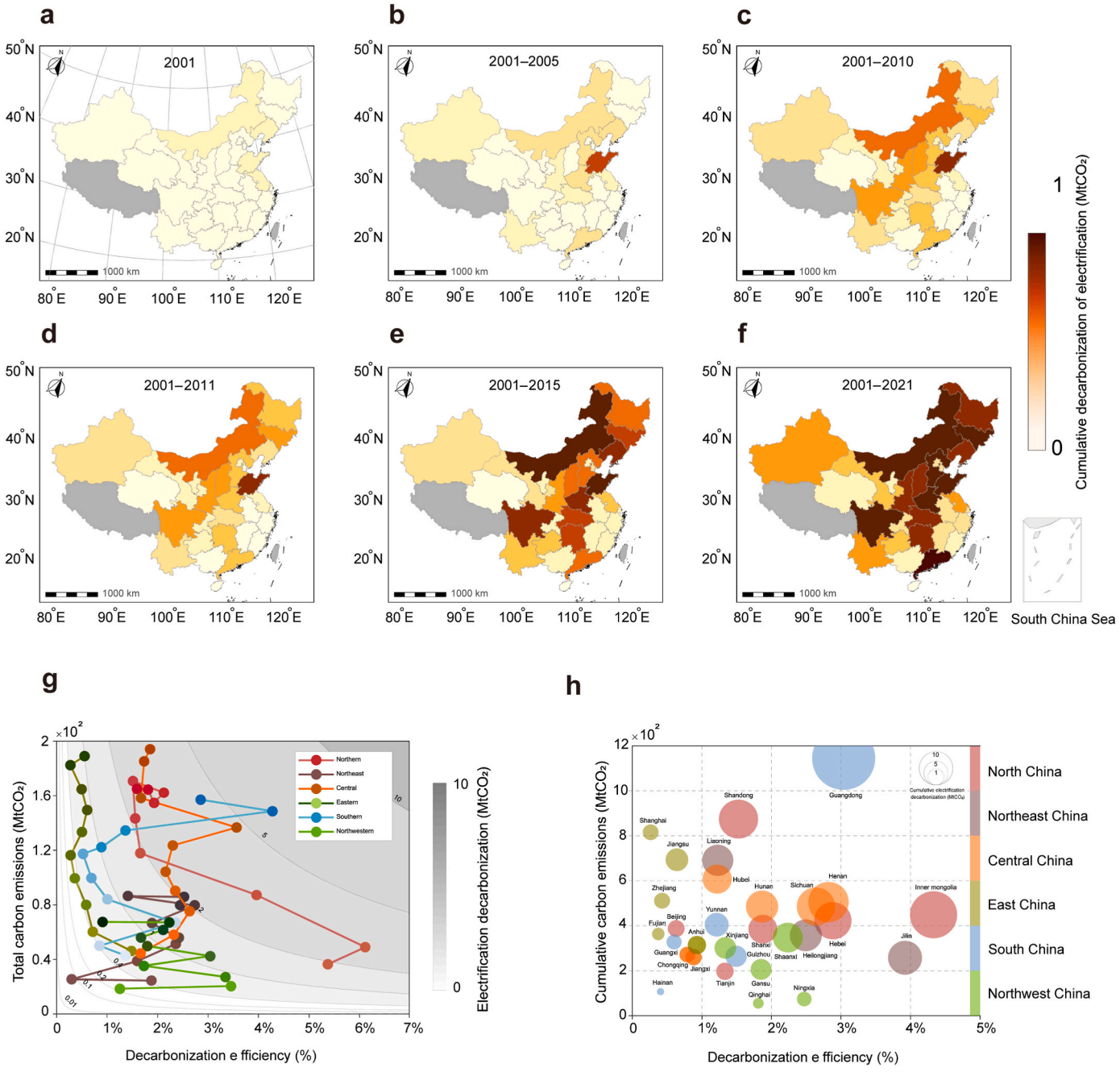


Fig. 6. Electrification-driven decarbonization of the transport sector in China (2000–2021): (a–f) Spatial and temporal variations in cumulative decarbonization of the transport sector in 30 provinces in China. The deepening color intensity reveals a clear geographical shift in decarbonization leadership from northern to southern provinces over time; (g) Decarbonization efficiency of the transport sector across grid regions; (h) Decarbonization efficiency of the transport sector across provinces. Notes: Points in (g) are color-coded on a light-to-dark gradient corresponding to biennial intervals from 2000 to 2021. The bubble size in (h) represents the cumulative carbon emissions, the y-axis shows the absolute decarbonization benefit from electrification, and the x-axis represents decarbonization efficiency.

the efficiency of the south China grid and the central China grid gradually increased, with the south China grid achieving the highest efficiency. Fig. 6 h further analyzes the decarbonization potential of each province. Notably, Inner Mongolia (4.6 %) and Jilin (4.2 %) have relatively high decarbonization efficiencies, with Inner Mongolia's cumulative decarbonization reaching 19.5 $MtCO_2$, primarily due to the extensive use of wind power. In contrast, provinces such as Shanghai (0.3 %), Hainan (0.4 %), and Guangxi (0.6 %) perform poorly in terms of decarbonization efficiency. The reasons for this poor performance vary. For instance, Shanghai's developed economy results in significant carbon emissions from the transportation sector, with a cumulative emission of 8.3 $MtCO_2$ over the past two decades. Meanwhile, Hainan and Guangxi have a cumulative emission of only 54 % (4.4 $MtCO_2$) of Shanghai's emissions over the same period. The lower decarbonization efficiency in these provinces are mainly due to the lower amount of electrification decarbonization, indicating substantial room for

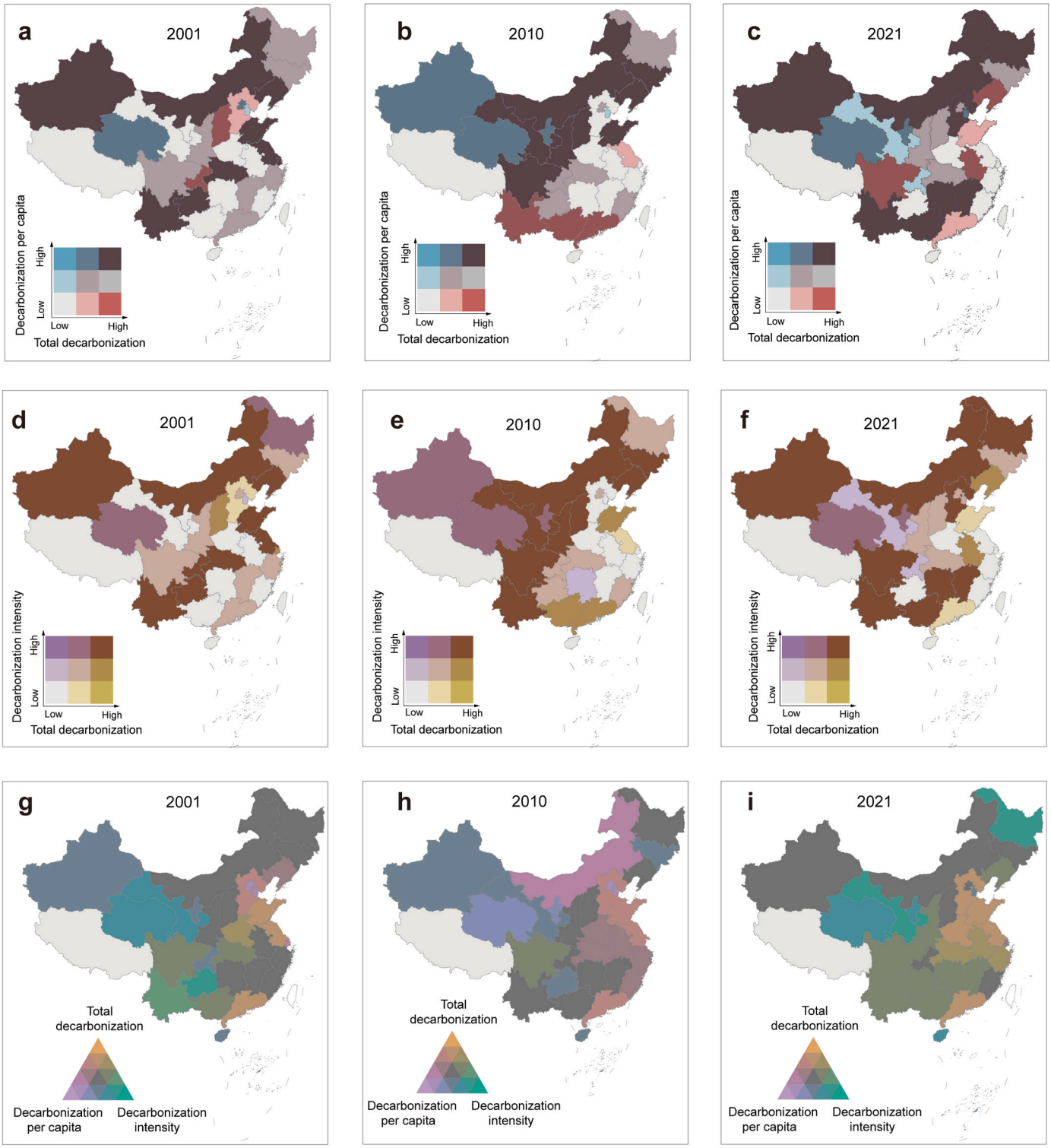


Fig. 7. Multi-scale assessment of provincial electrification-driven decarbonization in China's transport sector: (a–c) Decarbonization per capita; (d–f) Decarbonization intensity; (g–i) Total decarbonization.

improvement in the decarbonization effect of electrification.

Moreover, considering the differences in economic aggregates, resource endowments, and transportation demand among provinces, Fig. 7 provides a detailed analysis and comparison of decarbonization levels in the electrification of the transportation sector across different emission scales. As illustrated in Fig. 7, there are significant differences in assessment results across different emission scales, highlighting regional heterogeneity. Fig. 7 a illustrates the annual electrification-driven decarbonization performance across provinces, evaluating total decarbonization, decarbonization intensity, and decarbonization per capita. Guangdong province leads in electrification-driven decarbonization, achieving 1.7 $MtCO_2$ annually, followed by Inner Mongolia with 0.9 $MtCO_2$. Inner Mongolia also excels in decarbonization per capita at 38.1 $kgCO_2$ per capita, with Jilin and Guangdong following at 18.4 $kgCO_2$ and 13.9 $kgCO_2$ per capita, respectively. Additionally, Inner Mongolia demonstrates the highest decarbonization intensity, averaging 3477.7 $kgCO_2$ per 1000 yuan annually, with Sichuan province next at 2382.6 kg per 1000 yuan from 2000 to 2021. This performance is primarily attributed to Inner Mongolia's clean energy structure, leading the nation in clean energy generation, and Sichuan's extensive hydropower resources, which account for over 80 % of its electricity supply. The significant contribution of clean energy reduces carbon emissions from electrified transport, thereby increasing decarbonization intensity.

Conversely, the five lowest-performing provinces in terms of decarbonization intensity are Fujian (195.7 $kgCO_2$ per 1000 yuan), Zhejiang (261.1 $kgCO_2$ per 1000 yuan), Shanghai (300.3 $kgCO_2$ per 1000 yuan), Jiangsu (332.5 kg per 1000 yuan), and Anhui (347.1 kg per 1000 yuan), all located within the east China grid region. Their poor performance is largely due to high economic development, substantial demand for electric energy, a high proportion of oil and coal consumption in their transportation energy structure, and insufficient access to clean energy. Fig. 7a–c comprehensively examine the decarbonization performance in each province from the perspectives of per capita decarbonization and total decarbonization. It is observed that prior to 2010, the northern regions outperformed the southern regions. However, after 2010, the southern regions showed signs of improvement, particularly with the rise in per capita decarbonization and total decarbonization in Yunnan, Hunan, Guangxi, and Jiangxi. Notably, by 2021, certain provinces such as Inner Mongolia, Yunnan, and Hunan demonstrated excellent performance in both total and per capita decarbonization. In contrast, several economically advanced coastal provinces, including Jiangsu, Zhejiang, Shanghai, and Fujian, ranked at the bottom in these categories. Fig. 7d–f further assess the electrification-driven decarbonization performance in terms of decarbonization intensity and total decarbonization. provinces within the northwest China power grid region, with an average annual total of 0.9 $MtCO_2$ and an intensity of 1831.8 $kgCO_2$ per 1000 yuan, exhibited the best carbon emission intensity performance. Conversely, several provinces in the east China power grid region, with an average annual total of 2.8 $MtCO_2$ and an intensity of 1351.3 $kgCO_2$ per 1000 yuan, had the poorest performance. This disparity is primarily attributed to the abundant renewable energy resources in the northwest, such as wind and solar power, and the relatively lower average proportion of oil and coal consumption (85.3 %) compared to other provinces. The higher dependence on fossil fuels in the east China power grid region, with an average oil and coal consumption proportion of 95.5 %, is also a contributing factor.

5.2. Deep-dive case studies: contrasting decarbonization pathways in Guangdong and Inner Mongolia

Our national-level results reveal significant heterogeneity in decarbonization efficiency across provinces. To elucidate the underlying drivers of this variation, we selected Guangdong and Inner Mongolia for a comparative case study (Table 2). Guangdong represents an economically advanced, export-oriented coastal province with high transport demand and rapid electric vehicle (EV) penetration, but a historically carbon-intensive power grid. In contrast, Inner Mongolia is a resource-rich northern province with one of the highest shares of renewable energy in its grid mix in the nation, positioning it as a key clean energy hub. This stark contrast allows for a controlled comparison of how electrification benefits are mediated by regional economic structure and, most critically, power system characteristics.

Table 2
Comparative analysis of key covariates for Guangdong and Inner Mongolia.

Covariate	Guangdong	Inner Mongolia	Data Source
GDP (Billion yuan)	~12,400	~2158	National Bureau of Statistics [44]
NEV penetration rate (%)	High	Moderate	Our data, CPCA [45]
Public chargers (Total)	>667,250 (by 2024)	~32,551 (by 2024)	EVCIPA [46]
Renewable share in power mix (%)	Low (<30 %)	High (>50 %)	[47]
Key policy focus	NEV manufacturing, exports	"Solar + Storage + Charging", Green H_2	[48, 49]
Decarbonization efficiency (%)	Low	High	Our Results

Notes: GDP data are for 2021. NEV penetration rates are estimates based on available data. Public charger data as of the end of 2024. Renewable share varies annually; representative figures shown. Policy focus summarizes key provincial documents.

Our analysis reveals a "policy paradox": Guangdong, despite leading the nation in absolute decarbonization volume and EV infrastructure deployment, exhibits lower decarbonization efficiency than Inner Mongolia. The reason lies in the critical interplay between transport and power sectors. In Guangdong, the massive deployment of EVs was concurrent with significant growth in fossil fuel-based power generation needed to meet soaring industrial and residential demand, thus diluting the climate benefits of each EV.

Conversely, Inner Mongolia's success in achieving high decarbonization efficiency stems from its strategic coupling of renewable energy development with transport electrification. As a national renewable energy base, the province has proactively implemented policies to promote integrated "solar, storage, and charging" projects, leveraging its abundant wind and solar resources to ensure that the electricity powering its growing EV fleet is progressively cleaner. This synergy multiplies the climate benefit of each EV added to its fleet.

This comparative analysis provides a powerful, evidence-based explanation for the patterns observed in our main results. It demonstrates unequivocally that grid carbon intensity is a more powerful determinant of decarbonization efficiency than the rate of EV adoption alone. This finding underscores the necessity of synchronizing transport and power sector policies to maximize the climate returns on electrification investments.

5.3. Methodological robustness and contribution

Uncertainty analysis constitutes a critical component of the decarbonization methodological framework developed in this work. For this study, estimation uncertainties primarily arise from the influence of adjustable parameter configurations in computational procedures on numerical integration outcomes. To quantify this, we conducted multiple independent numerical experiments across 30 provinces, iterating adjustable parameters within the range $[1, 10^6]$. As illustrated in Fig. 8, convergence curves reveal that provincial decomposition results stabilize progressively with parameter variations, while relative errors asymptotically approach zero. At parameter values exceeding 16,000, the model achieves a relative error below 0.01 %, confirming the numerical robustness and stability of our algorithmic framework.

Beyond its numerical stability, the DSD framework's primary

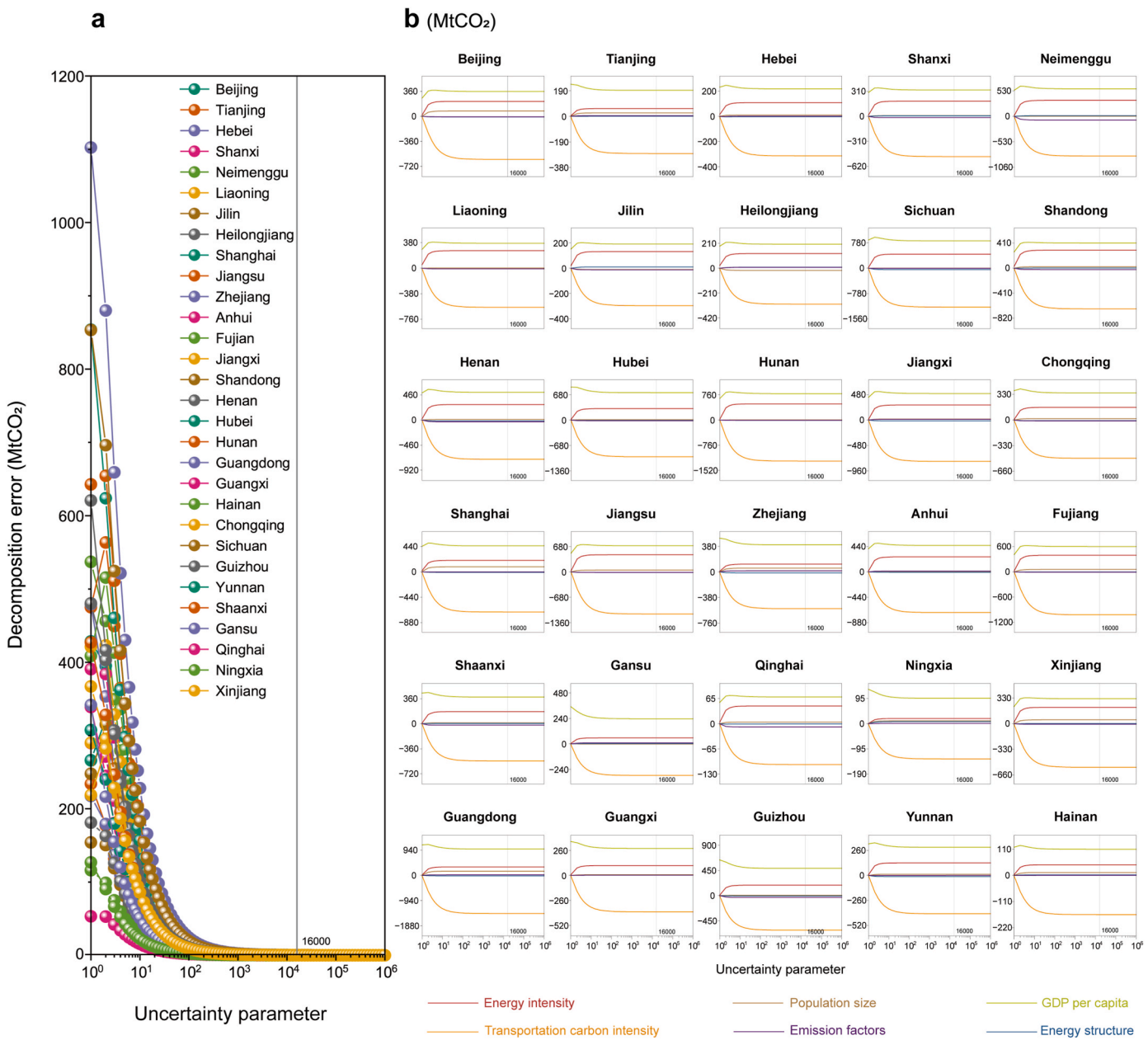


Fig. 8. Uncertainty testing of the model framework: (a) Iterative plots of model error with adjustable parameters; (b) Iterative plots of model results with adjustable parameters.

contribution lies in the unique granularity of its outputs, which provides insights that are not recoverable using conventional methods like LMDI. As highlighted in the literature review, a conventional LMDI analysis would have conflated the complex shifts within the energy mix into a single, aggregated "energy structure effect." This would have masked the critical underlying trade-off between the emissions reduction from displacing fossil fuels and the emissions increase from growing electricity demand. The DSD method, by resolving this unit-sum constraint, enabled us to construct our decarbonization assessment model and precisely isolate three distinct mechanisms: the electricity substitution effect, the electricity demand growth effect, and the grid decarbonization effect.

This granular decomposition yields domain knowledge that would otherwise be obscured. For instance, our finding that southern provinces exhibit high absolute decarbonization but low efficiency is a nuanced conclusion only achievable by simultaneously quantifying these three effects. An LMDI analysis might have simply shown a positive "structural effect" from electrification, failing to reveal that a strong substitution effect was being significantly counteracted by a weak or even negative grid decarbonization effect. The DSD framework thus provides a superior diagnostic toolkit. It allows policymakers to pinpoint why decarbonization is underperforming in a given province: is the bottleneck the slow pace of EV adoption (a weak substitution effect) or the carbon intensity of the power supply (a poor grid decarbonization effect)? By answering this question, the DSD-based framework provides a direct scientific basis for tailoring the targeted and strengthens the responses to the three research questions posed in Section 1.

5.4. Policy implications for deep decarbonization

The empirical evidence presented in this study offers several critical insights for crafting the next generation of transport decarbonization policies in China. As the era of nascent market cultivation gives way to a phase of accelerated mass adoption of electric vehicles, the policy focus

must evolve from simply promoting electrification to strategically managing a systemic, multi-sector energy transition. To this end, our findings support a move towards a more sophisticated and evidence-based policy architecture, structured around the following interventions.

(a) Implementing differentiated provincial decarbonization policies

Our findings reveal pronounced disparities in decarbonization performance across Chinese provinces, which renders a uniform, "one-size-fits-all" national policy approach inefficient. A pragmatic policy recommendation is the urgent transition toward differentiated, province-specific incentive and investment structures. For economically advanced coastal provinces such as Guangdong, our data indicate high absolute decarbonization volumes, yet suboptimal efficiency due to carbon-intensive power grids. Thus, policy emphasis must be placed on aggressively decarbonizing local electricity supply—by accelerating the deployment of offshore wind and solar PV generation, alongside investing in inter-provincial ultra-high-voltage transmission to import cleaner energy. Simultaneously, these regions should implement grid impact management strategies such as time-of-use electricity pricing, smart charging incentives, and vehicle-to-grid (V2G) applications to alleviate peak load stress. Conversely, in provinces endowed with abundant clean energy resources, such as Inner Mongolia and Sichuan, where decarbonization efficiency is notably higher, policies should focus on reinforcing their role as clean energy hubs through enhanced inter-provincial transmission infrastructure. Meanwhile, for provinces lagging in both absolute decarbonization benefits and efficiency, the immediate priority should be foundational capacity building. This entails accelerating the rollout of public charging infrastructure to mitigate range anxiety, and prioritizing the full electrification of public vehicle fleets (e.g., buses, taxis, municipal vehicles)—offering an effective and economically viable starting point for their low-carbon transition.

(b) Mandating system-wide coordination of transport and power sectors

Our study empirically confirms that lagging grid decarbonization is a critical bottleneck that can severely erode or even negate the climate benefits of transport electrification. This points to a systemic coordination failure that requires an institutional solution beyond general calls for cooperation. A key policy recommendation is therefore to mandate synchronized planning between the transport and power sectors, particularly at the regional level. Future national development frameworks, such as the forthcoming 15th Five-Year Plan, should consider establishing binding, joint decarbonization targets for the transport and power sectors combined, allocated to regional grid authorities. Such a mechanism would compel provincial governments within a grid region to engage in integrated resource planning, ensuring that new renewable energy capacity is developed in lockstep with projected growth in electric vehicle fleets and charging demand, thereby closing the gap between demand-side and supply-side transitions [50].

(c) Integrating transport decarbonization with economic restructuring

Our decomposition analysis revealed that while economic growth remains the primary driver of emissions, the most powerful decarbonization lever over the past two decades has been the reduction in "transport demand intensity"—the amount of transport activity required per unit of GDP. This finding highlights the profound importance of embedding transport policy within a broader strategy of economic restructuring. Rather than focusing solely on technological substitution within the transport sector, a more impactful, long-term approach is to actively foster economic models that are inherently less reliant on the physical movement of goods. Industrial and urban planning policies should therefore prioritize and incentivize the growth of the digital services economy [51], promote flexible work arrangements [52], and invest in intelligent logistics platforms that optimize freight efficiency [53]. By doing so, policymakers can foster a progressive decoupling of economic prosperity from transport emissions, addressing the root driver of carbon growth.

(d) Leveraging advanced transportation systems for systemic efficiency

While electrification is a cornerstone of decarbonization, it should not be the sole focus. Our findings on the significant role of energy intensity suggest that improving the overall efficiency of the transport system is equally crucial. With advancements in green and energy-efficient technologies and the development of the Internet, it is imperative to harness the mitigation potential of advanced transportation systems by integrating energy management strategies [54], energy-efficient heterogeneous networks [55], advanced driver assistance systems [56], electric-automated mobility-on-demand [57], and graph-based multi-modal one-to-many ride-matching [58] to improve energy efficiency and reduce carbon emissions.

Overall, our study reviews and evaluates the status of transport electrification and decarbonization initiatives in China. Building on the empirical evidence presented, the analysis proposes spatially differentiated and ambitious deep-decarbonization strategies to accelerate provincial-level transitions toward net-zero emissions. These findings provide a comprehensive resolution to research question 3 posed in Section 1.

6. Conclusion

6.1. Key findings

Our work develops an integrated top-down and bottom-up analytical framework to assess carbon emission patterns and electrification-driven decarbonization benefits across China's 30 provinces. The findings reveal a critical policy paradox: while transportation electrification has achieved measurable cumulative decarbonization of 210.4 $MtCO_2$ by 2021, these gains have been substantially offset by persistent emission growth driven by economic expansion and, crucially, by indirect emissions from a power grid still in transition. This underscores the vital conclusion that electrification, as a standalone strategy, is insufficient to deliver the deep decarbonization required to meet climate targets.

The primary contribution of this study lies in its provincial-level diagnostic toolkit, which moves beyond aggregate national analyses to uncover significant regional heterogeneity in decarbonization performance. We demonstrate that while southern provinces lead in absolute decarbonization from electrification, their decarbonization efficiency lags behind that of northern regions. This inefficiency highlights a critical disconnect between the pace of vehicle electrification and the progress of grid decarbonization. The key implication for policymakers is that a one-size-fits-all national electrification mandate is suboptimal. Instead, region-specific strategies are imperative, requiring the harmonization of transportation policies with power sector planning to ensure that rising electricity demand from transport is met with increasingly clean energy sources. Our framework provides the quantitative basis for tailoring such coordinated interventions.

6.2. Limitations and future work

A key limitation of this study lies in its reliance on aggregated data from the "Transport, Storage, and Post" sector, as constrained by the structure of official statistics. This approach hampers a more granular analysis of emission drivers and decarbonization benefits across specific transport modes—such as road, rail, waterway, and aviation. Given that road transport represents the primary target for electrification, the use

of such aggregated data may underestimate or obscure the true policy impact on this critical segment. Furthermore, constrained by the release cycles of provincial energy statistics, the temporal scope of this study concludes with the year 2021. Subsequent analyses, incorporating newly released data in the coming years, will be essential to quantify the decarbonization benefits and system-wide carbon emission impacts during this phase of accelerated electrification.

Future research should prioritize the integration of emerging technological pathways—such as green hydrogen, advanced battery systems, and the systemic potential of vehicle-to-grid (V2G) integration—into provincial-level modeling frameworks. Incorporating these disruptive technologies is critical for developing comprehensive and effective carbon-neutrality roadmaps. Moreover, refining the modeling of decarbonization trajectories for China's transport sector—particularly its rapidly growing road transport segment—remains imperative. Follow-up studies should therefore evaluate the feasibility of decarbonization targets under diverse electrification scenarios, align provincial carbon budgets with national "dual carbon" objectives, and help bridge the gap between China's carbon-neutrality pledges and sector-wide implementation. Such alignment is vital to ensure the compatibility of transport decarbonization efforts with global climate goals.

CRediT authorship contribution statement

Xiwang Xiang: Writing – original draft, Visualization, Methodology. **Tianduo Peng:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization. **Ershun Du:** Writing – review & editing, Validation, Supervision. **Chenjun Hou:** Visualization, Validation, Data curation. **Qin Wang:** Supervision, Writing – review & editing.

Declaration of competing interest

no conflcit of interest statement.

Acknowledgment

This work was supported by Carbon Neutrality and Energy System Transformation (CNEST) Program, the National Top Think-Tank Research Center of China National Petroleum Corp (No. 2025-ZL-33) and the Think-Tank Center of Tsinghua University (No. 20258060010).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.etran.2025.100529.

Data availability

Data will be made available on request.

References

[1] Rogelj J, Luderer G, Pietzcker RC, et al. Energy system transformations for limiting end-of-century warming to below 1.5 C. Nat Clim Change 2015;5(6):519–27.
[2] Traver M, Ebbinghaus A, Moljord K, Morganti K, Pearson R, Vermeire M. Fuels and transportation. Energy transition: climate action and circularity. ACS Publications; 2022. p. 83–129.
[3] Sun Y, Yang Y, Liu S, Li Q. Research on transportation carbon emission peak prediction and judgment system in China. Sustainability 2023;15(20):14880.
[4] Asif M, Muneer T. Energy supply, its demand and security issues for developed and emerging economies. Renew Sustain Energy Rev 2007;11(7):1388–413. https://doi.org/10.1016/j.rser.2005.12.004 [published Online First: Epub Date].
[5] Bistline JE, Blanford G, Grant J, et al. Economy-wide evaluation of CO2 and air quality impacts of electrification in the United States. Nat Commun 2022;13(1):6693.
[6] Yuan M, Thellufsen JZ, Lund H, Liang Y. The electrification of transportation in energy transition. Energy 2021;236:121564. https://doi.org/10.1016/j.energy.2021.121564 [published Online First: Epub Date].
[7] Bogdanov D, Ram M, Khalili S, Aghahosseini A, Fasihi M, Breyer C. Effects of direct and indirect electrification on transport energy demand during the energy transition. Energy Policy 2024;192:114205. https://doi.org/10.1016/j.enpol.2024.114205 [published Online First: Epub Date]|.
[8] Zhang R, Hanaoka T. Cross-cutting scenarios and strategies for designing decarbonization pathways in the transport sector toward carbon neutrality. Nat Commun 2022;13(1):3629. https://doi.org/10.1038/s41467-022-31354-9 [published Online First: Epub Date]|.
[9] van Heerden R, Edelenbosch OY, Daioglou V, et al. Demand-side strategies enable rapid and deep cuts in buildings and transport emissions to 2050. Nat Energy 2025;10(3):380–94. https://doi.org/10.1038/s41560-025-01703-1 [published Online First: Epub Date]|.
[10] Edelenbosch OY, van Vuuren DP, Blok K, Calvin K, Fujimori S. Mitigating energy demand sector emissions: the integrated modelling perspective. Appl Energy 2020;261:114347. https://doi.org/10.1016/j.apenergy.2019.114347 [published Online First: Epub Date]|.
[11] Andersson Ö, Börjesson P. The greenhouse gas emissions of an electrified vehicle combined with renewable fuels: life cycle assessment and policy implications. Appl Energy 2021;289:116621.
[12] Creutzig F, Simoes SG, Leipold S, et al. Demand-side strategies key for mitigating material impacts of energy transitions. Nat Clim Change 2024;14(6):561–72.
[13] Anandarajah G, McDowall W, Ekins P. Decarbonising road transport with hydrogen and electricity: long term global technology learning scenarios. Int J Hydrogen Energy 2013;38(8):3419–32.
[14] Rapson D, Muehlegger E. Global transportation decarbonization. J Econ Perspect 2023;37(3):163–88.
[15] Zhao J, Zhang Y, He X, Axmacher JC, Sang W. Gains in China's sustainability by decoupling economic growth from energy use. J Clean Prod 2024;448:141765. https://doi.org/10.1016/j.jclepro.2024.141765 [published Online First: Epub Date]|.
[16] Bistline JET, Blanford G, Grant J, et al. Economy-wide evaluation of CO2 and air quality impacts of electrification in the United States. Nat Commun 2022;13(1):6693. https://doi.org/10.1038/s41467-022-33902-9 [published Online First: Epub Date]|.
[17] Xiang X, Ma X, Ma Z, Ma M. Operational carbon change in commercial buildings under the carbon neutral goal: a LASSO–WOA approach. Buildings 2022;12(1):54.
[18] Sorrell S, Miller R, Bentley R, Speirs J. Oil futures: a comparison of global supply forecasts. Energy Policy 2010;38(9):4990–5003.
[19] Shafiei E, Davidsdottir B, Leaver J, Stefansson H, Asgeirsson EI. Energy, economic, and mitigation cost implications of transition toward a carbon-neutral transport sector: a simulation-based comparison between hydrogen and electricity. J Clean Prod 2017;141:237–47.
[20] Fortes P, Simoes SG, Gouveia JP, Seixas J. Electricity, the silver bullet for the deep decarbonisation of the energy system? cost-effectiveness analysis for Portugal. Appl Energy 2019;237:292–303.
[21] Needell ZA, McNerney J, Chang MT, Trancik JE. Potential for widespread electrification of personal vehicle travel in the United States. Nat Energy 2016;1(9):1–7.
[22] Holland SP, Kotchen MJ, Mansur ET, Yates AJ. Why marginal CO2 emissions are not decreasing for US electricity: estimates and implications for climate policy. Proc Natl Acad Sci 2022;119(8):e2116632119.
[23] Zhou X-Y, Xu Z, Zheng J, et al. Internal spillover effect of carbon emission between transportation sectors and electricity generation sectors. Renew Energy 2023;208:356–66.
[24] Xiang X, Ma M, Ma X, et al. Historical decarbonization of global commercial building operations in the 21st century. Appl Energy 2022;322:119401.
[25] Xiang X, Zhou N, Ma M, Feng W, Yan R. Global transition of operational carbon in residential buildings since the millennium. Adv Appl Energy 2023;11:100145.
[26] Köne AÇ, Büke T. Factor analysis of projected carbon dioxide emissions according to the IPCC based sustainable emission scenario in Turkey. Renew Energy 2019;133:914–8.
[27] Liu M, Zhang X, Zhang M, et al. Influencing factors of carbon emissions in transportation industry based on CD function and LMDI decomposition model: china as an example. Environ Impact Assess Rev 2021;90:106623. https://doi.org/10.1016/j.eiar.2021.106623 [published Online First: Epub Date]|.
[28] Yu Y, Li S, Sun H, Taghizadeh-Hesary F. Energy carbon emission reduction of China's transportation sector: an input–output approach. Econ Anal Pol 2021;69:378–93.
[29] Ma J-J, Du G, Xie B-C. CO2 emission changes of China's power generation system: input-output subsystem analysis. Energy Policy 2019;124:1–12.
[30] Zhang X, Liu K, Zhang Z. Life cycle carbon emissions of two residential buildings in China: Comparison and uncertainty analysis of different assessment methods. J Clean Prod 2020;266:122037.
[31] Dey S, Caulfield B, Ghosh B. Modelling uncertainty of vehicular emissions inventory: a case study of Ireland. J Clean Prod 2019;213:1115–26. https://doi.org/10.1016/j.jclepro.2018.12.125 [published Online First: Epub Date]|.
[32] Bishop JD, Molden N, Boies AM. Using portable emissions measurement systems (PEMS) to derive more accurate estimates of fuel use and nitrogen oxides emissions from modern Euro 6 passenger cars under real-world driving conditions. Appl Energy 2019;242:942–73.
[33] Raza MY, Lin B. Energy transition, carbon trade and sustainable electricity generation in Pakistan. Appl Energy 2024;372:123782.
[34] Vaninsky A. Factorial decomposition of CO2 emissions: a generalized Divisia index approach. Energy Econ 2014;45:389–400.

[35] Boratyński J. Decomposing structural decomposition: the role of changes in individual industry shares. Energy Econ 2021;103:105587. https://doi.org/10.1016/j.eneco.2021.105587 [published Online First: Epub Date]|.
[36] Jing H, Chen Y, Ma M, Feng W, Xiang X. Global carbon transition in the passenger transportation sector over 2000–2021. Sustain Prod Consum 2024;51:556–71. https://doi.org/10.1016/j.spc.2024.10.006 [published Online First: Epub Date]|.
[37] Jing H, Chen Y, Ma M, Feng W, Xiang X. Unlocking global freight transport decarbonization potential: role of modal shifts. Transport Res Transport Environ 2025;148:105025. https://doi.org/10.1016/j.trd.2025.105025 [published Online First: Epub Date]|.
[38] Chen R, Xu P, Chen L, Yao H. Did electrification of the building sector achieve carbon mitigation? A provincial retrospection in China. Build Environ 2024;248:111084.
[39] Shan Y, Guan D, Zheng H, et al. China CO2 emission accounts 1997–2015. Sci Data 2018;5(1):170201. https://doi.org/10.1038/sdata.2017.201 [published Online First: Epub Date]|.
[40] Song Z. Economic growth and carbon emissions: estimation of a panel threshold model for the transition process in China. J Clean Prod 2021;278:123773. https://doi.org/10.1016/j.jclepro.2020.123773 [published Online First: Epub Date]|.
[41] Yu Y, Dai C, Wei Y, Ren H, Zhou J. Air pollution prevention and control action plan substantially reduced PM2.5 concentration in China. Energy Econ 2022;113:106206. https://doi.org/10.1016/j.eneco.2022.106206 [published Online First: Epub Date]|.
[42] Zhao M, Sun T. Dynamic spatial spillover effect of new energy vehicle industry policies on carbon emission of transportation sector in China. Energy Policy 2022;165:112991. https://doi.org/10.1016/j.enpol.2022.112991 [published Online First: Epub Date]|.
[43] Mi Z, Meng J, Guan D, et al. Chinese CO2 emission flows have reversed since the global financial crisis. Nat Commun 2017;8(1). https://doi.org/10.1038/s41467-017-01820-w [published Online First: Epub Date]|.
[44] Nbo Statistics. Gross regional product. 2025.
[45] Liang Y, Lu H. Dynamic evaluation and regional differences analysis of the NEV industry development in China. Sustainability 2022;14(21):13864. https://doi.org/10.3390/su142113864 [published Online First: Epub Date]|.
[46] EVCIPA. Analysis of the growth of national public charging infrastructure in the first quarter of 2025. Secondary analysis of the growth of national public charging infrastructure in the first quarter of 2025. 2025. https://www.evcipa.org.cn/newsinfo/8310587.html.
[47] Zhu Z, Ma S, Mao H, et al. Spatial planning for China's renewable energy expansion toward 2030. Appl Energy 2025;398:126413. https://doi.org/10.1016/j.apenergy.2025.126413 [published Online First: Epub Date]|.
[48] Li Y, Huang Y, Liang Y, Song C, Liao S. Economic and carbon reduction potential assessment of vehicle-to-grid development in guangdong province. Energy 2024;302:131742. https://doi.org/10.1016/j.energy.2024.131742 [published Online First: Epub Date]|.
[49] Liu W, Hu W, Lund H, Chen Z. Electric vehicles and large-scale integration of wind power – the case of Inner Mongolia in China. Appl Energy 2013;104:445–56. https://doi.org/10.1016/j.apenergy.2012.11.003 [published Online First: Epub Date]|.
[50] Zhang C, Lu X, Chen S, et al. Synergies of variable renewable energy and electric vehicle battery swapping stations: case study for Beijing. eTransportation 2024;22:100363. https://doi.org/10.1016/j.etran.2024.100363 [published Online First: Epub Date]|.
[51] Nie S, Cao X, Li Z, Liu M, Zhang Y. Supply chain digitization in the net-zero era: the impact of digital technology, renewable energy, and infrastructure. Energy Econ 2025;144:108403. https://doi.org/10.1016/j.eneco.2025.108403 [published Online First: Epub Date]|.
[52] Zhang R, Zhang J. Long-term pathways to deep decarbonization of the transport sector in the post-COVID world. Transp Policy 2021;110:28–36. https://doi.org/10.1016/j.tranpol.2021.05.018 [published Online First: Epub Date]|.
[53] Yin Y, Wang H, Deng X. Real-time logistics transport emission monitoring-integrating artificial intelligence and internet of things. Transport Res Transport Environ 2024;136:104426. https://doi.org/10.1016/j.trd.2024.104426 [published Online First: Epub Date]|.
[54] Gan J, Li S, Wei C, Deng L, Tang X. Intelligent learning algorithm and intelligent transportation-based energy management strategies for hybrid electric vehicles: a review. IEEE Trans Intell Transport Syst 2023;24(10):10345–61.
[55] Jiang D, Huo L, Zhang P, Lv Z. Energy-efficient heterogeneous networking for electric vehicles networks in smart future cities. IEEE Trans Intell Transport Syst 2020;22(3):1868–80.
[56] Antony MM, Whenish R. Advanced driver assistance systems (ADAS). Automotive embedded systems: key technologies, innovations, and applications. Springer; 2021. p. 165–81.
[57] Sheng Y, Lin Y, Zeng H, Yu Y, Guo Q, Xie S. Emission-concerned coordinated dispatching of electrified autonomous mobility-on-demand system and power system incorporating heterogeneous spatiotemporal scales. Sustain Cities Soc 2023;98:104755.
[58] Meshkani SM, Farooq B. A generalized ride-matching approach for sustainable shared mobility. Sustain Cities Soc 2022;76:103383.